

Q1 A basic feasible solution of an $m \times n$ transportation problem is said to be non-degenerate, if basic feasible solution contains exactly----- number of individual allocation in----- positions.

- (A) $m + n + 1$, independent
 (B) $m + n - 1$, independent
 (C) $m + n - 1$, appropriate
 (D) $m - n + 1$, independent

Q2 Consider the following Linear programming problem (LPP) :

Maximize $z = x_1 + x_2$

Subject to the constraints:

$$x_1 + 2x_2 \leq 2000$$

$$x_1 + x_2 \leq 1500$$

$$x_2 \leq 600$$

$$\text{and } x_1, x_2 \geq 0$$

The solution of the above LPP is:

- (A) $x_1 = 750, x_2 = 750, z = 1500$
 (B) $x_1 = 500, x_2 = 1000, z = 1500$
 (C) $x_1 = 1000, x_2 = 500, z = 1500$
 (D) $x_1 = 900, x_2 = 600, z = 1500$

Q3 The Boolean expression

$AB + A\bar{B} + \bar{A}C + AC$ is unaffected by the value of the Boolean variable -----

- (A) A (B) B
 (C) C (D) A, B and C

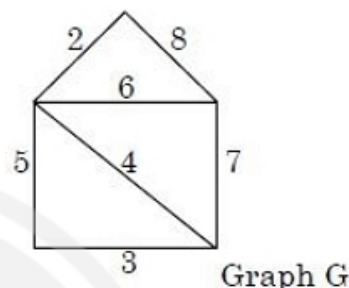
Q4 What are the greatest lower bound (GLB) and the least upper bound (LUB) of the sets $A = \{3, 9, 12\}$ and $B = \{1, 2, 4, 5, 10\}$ if they exist in poset $(z^+, /)$?

- (A) $A(\text{GLB} - 3, \text{LUB} - 36)$
 : $B(\text{GLB} - 1, \text{LUB} - 20)$
 (B) $A(\text{GLB} - 3, \text{LUB} - 12)$;
 $B(\text{GLB} - 1, \text{LUB} - 10)$
 (C) $A(\text{GLB} - 1, \text{LUB} - 36)$;
 $B(\text{GLB} - 2, \text{LUB} - 20)$
 (D) $A(\text{GLB} - 1, \text{LUB} - 12)$;
 $B(\text{GLB} - 2, \text{LUB} - 10)$

Q5 Let (P) be the set of all people. Let (R) be a binary relation on (P) such that (a, b) is in R if a is a brother of b. Is (R) symmetric transitive, an equivalence relation, a partial order relation?

- (A) NO,NO,NO,NO (B) NO,NO,YES,NO
 (C) NO,YES,NO,NO (D) NO,YES,YES,NO

Q6 The weight of minimum spanning tree in graph G, calculated using Kruskal's algorithm is :



- (A) 14 (B) 15
 (C) 17 (D) 18

Q7 A tree has $2n$ vertices of degree 1, $3n$ vertices of degree 2, n vertices of degree 3. Determine the number of vertices and edges in tree.

- (A) 12,11 (B) 11,12
 (C) 10,11 (D) 9,10

Q8 How many reflexive relations are there on a set with 4 elements?

- (A) 2^4 (B) 2^{12}
 (C) 4^2 (D) 2

Q9 A non-pipelined system takes 30 ns to process a task. The same task can be processed in a four-segment pipeline with a clock cycle of 10 ns. Determine the speed up of the pipeline for 100 tasks.

- (A) 3 (B) 4
 (C) 3.91 (D) 2.91

Q10 A computer uses a memory unit of 512 K words of 32 bits each. A binary instruction code is stored in one word of the memory. The instruction has four parts: an addressing mode field to specify one of the two-addressing mode (direct and indirect), an operation code, a register



code part to specify one of the 256 registers and an address part. How many bits are there in addressing mode part, opcode part, register code part and the address part?

- (A) 1, 3, 9, 19 (B) 1, 4, 9, 18
(C) 1, 4, 8, 19 (D) 1, 3, 8, 20

Q11 A micro instruction format has microoperation field which is divided into 2 subfields F1 and F2, each having 15 distinct microoperations, condition field CD for four status bits, branch field B R having four options used in conjunction with address field AD . The address space is of 128 memory words. The size of micro instruction is :

- (A) 19 (B) 18
(C) 17 (D) 20

Q12 $(142)_b + (112)_{b-2} = (75)_8$, find base b.

- (A) 3 (B) 6
(C) 7 (D) 5

Q13 The following program is stored in the memory unit of the basic computer. Give the content of accumulator register in hexadecimal after the execution of the program.

Location	Instruction
010	CLA
011	ADD 016
012	BUN 014
013	HLT
014	AND 017
015	BUN 013
016	C1A5
017	93 C6

- (A) A1B4 (B) 81 B 4
(C) A184 (D) 8184

Q14 What is the output of following C program?

```
...
#include <stdio.h>
main()
{
    int i, j, x = 0;
    for (i=0; i<5; ++i)
```

```
    for (j=0; j<i; ++j)
    {
        x+=(i+j-1);
        break ;
    }
    printf ("%d", x);
    ...
```

- (A) 6 (B) 5
(C) 4 (D) 3

Q15 Let A be the base class in C++ and B be the derived class from A with protected inheritance. Which of the following statement is false for class B?

- (A) Member function of class B can access protected data of class A
(B) Member function of Class B can access public data of class A
(C) Member function of class B cannot access private data of class A
(D) Object of derived class B can access public base class data

Q16 Which tag is used to enclose any number of javascript statements in HTML document?

- (A) <code> (B) <script>
(C) <title> (D) <body>

Q17 A rectangle is bound by the lines $x = 0$; $y = 0$; $x = 5$ and $y = 3$. The line segment joining $(-1, 0)$ and $(4, 5)$, if clipped against this window will connect the points -----

- (A) (0, 1) and (3, 3)
(B) (0, 1) and (2, 3)
(C) (0, 1) and (4, 5)
(D) (0, 1) and (3, 5)

Q18 Which of the following algorithms is not used for line clipping?

- (A) Cohen-Sutherland algorithm
(B) Sutherland-Hodgeman algorithm
(C) Liang-Barsky algorithm
(D) Nicholl-Lee-Nicholl algorithm

Q19



If we want to resize a 1024×768 pixels image to one that is 640 pixels wide with the same aspect ratio, what would be the height of the resized image?

- (A) 420 Pixels (B) 460 Pixels
(C) 480 Pixels (D) 540 Pixels

Q20 Which of the component module of DBMS does rearrangement and possible ordering of operations, eliminate redundancy in query and use efficient algorithms and indexes during the execution of a query?

- (A) query compiler
(B) query optimizer
(C) Stored data manager
(D) Database processor

Q21 Given two tables $R1(x,y)$ and $R2(y,z)$ with 50 and 30 numbers of tuples respectively. Find the maximum number of tuples in the output of natural join between tables $R1$ and $R2$ i.e. $R1 \bowtie R2$.
(* - Natural Join)

- (A) 30 (B) 20
(C) 50 (D) 1500

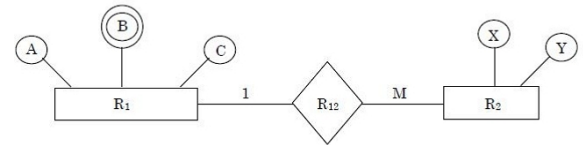
Q22 Given two tables EMPLOYEE (EID, ENAME, DEPTNO)
DEPARTMENT (DEPTNO, DEPTNAME)
Find the most appropriate statement of the given query :

Select count (*) 'total'
from EMPLOYEE
where DEPTNO IN (D1, D2)
group by DEPTNO
having count (*) > 5.

- (A) Total number of employees in each department D1 and D2
(B) Total number of employees of departments D1 and D2 if their total is > 5.
(C) Display total number of employees in both departments D1 and D2
(D) The output of the query must have atleast two rows

Q23 Find the minimum number of tables required for converting the following entity relationship

diagram into a relational database.



- (A) 2 (B) 4
(C) 3 (D) 5

Q24 A counting semaphore is initialized to 8. 3 wait () operations and 4 signal () operations are applied. Find the current value of the semaphore variable.

- (A) 9 (B) 5
(C) 1 (D) 4

Q25 Which of the following is not needed by an encryption algorithm used in Cryptography?

- (A) KEY (B) Message
(C) Ciphertext (D) User details

Q26 Which of the following CPU scheduling algorithms is/are supported by LINUX operating system?

- (A) Non-preemptive priority scheduling
(B) Preemptive priority scheduling and time-sharing CPU scheduling
(C) Time sharing scheduling only
(D) Priority scheduling only

Q27 Consider a paging system where translation looks aside buffer (TLB) a special type of associative memory is used with a hit ratio of 80%..

Assume that memory reference takes 80 nanoseconds and reference time to TLB is 20 nanoseconds. What will be the effective memory access time given an 80 % hit ratio?

- (A) 110 nanoseconds
(B) 116 nanoseconds
(C) 200 nanoseconds
(D) 100 nanoseconds

Q28 Suppose a system has 12 magnetic tape drives and at the time t_0 , three processes are allotted tape drives out of their need as given below:

Maximum Needs	Current Needs
------------------	------------------



p_0	10	5
p_1	4	2
p_2	9	2

At times t_0 , the system is in a safe state. Which of the following is a safe sequence so that deadlock is avoided?

- (A) $\langle p_0, p_1, p_2 \rangle$ (B) $\langle p_1, p_0, p_2 \rangle$
 (C) $\langle p_2, p_1, p_0 \rangle$ (D) $\langle p_0, p_2, p_1 \rangle$

- Q29** Which of the following interprocess communication model is used to exchange messages among co-operative processes?
 (A) Shared memory model
 (B) Message passing model
 (C) Shared memory and message passing model
 (D) Queues

- Q30** Given a CPU time slice of 2 ms and the following list of processes.

Process	Burst time (ms)	Arrival time
P1	3	0
P2	4	2
P3	5	5

Find average turnaround time and average waiting time using round-robin CPU scheduling.

- (A) 4,0 (B) 5.66,1.66
 (C) 5.66,0 (D) 7,2
- Q31** Which of the following methods are used to pass any number of parameters to the operating system through system calls?
 (A) Registers
 (B) Block or table in main memory
 (C) Stack
 (D) Block in main memory and stack
- Q32** Java Virtual Machine (JVM) is used to execute architectural neutral byte code. Which of the following is needed by the JVM for execution of Java Code?
 (A) Class loader only
 (B) Class loader and Java Interpreter
 (C) Class loader, Java Interpreter and API
 (D) Java Interpreter only

Q33

In a system for a restaurant, the main scenario for placing an order is given below :

- (a) Customer reads menu
 (b) Customer places order
 (c) Order is sent to the kitchen for preparation
 (d) Ordered items are served
 (e) Customer requests for a bill for the order
 (f) Bill is prepared for this order
 (g) Customer is given the bill
 (h) Customer pays the bill

A sequence diagram for the scenario will have at least how many objects among whom the messages will be exchanged.

- (A) 3 (B) 4
 (C) 5 (D) 6

- Q34** An ----- chart is a project schedule representation that presents project plan as a directed graph. The critical path is----- the sequence of -----tasks and it defines project-----
 (A) Activity, Shortest, Independent, Cost
 (B) Activity, Longest, Dependent, Duration
 (C) Activity, Longest, Independent, Duration
 (D) Activity, Shortest, Dependent, Duration

- Q35** Let $a^{2c} \bmod n = (a^c)^2 \bmod n$ and $a^{2c+1} \bmod n = a \cdot (a^c)^2 \bmod n$. For $a=7, b=17$, and $n=561$ what is the value of $a^b \bmod n$?
 (A) 160 (B) 166
 (C) 157 (D) 67

- Q36** A clique in an undirected graph $G = \langle V, E \rangle$ is a subset $V' \subseteq V$ of vertices, such that
 (A) If $(u, v) \in E$ then $u \in V'$ and $v \in V'$
 (B) If $(u, v) \in E$ then $u \in V'$ or $v \in V'$
 (C) Each pair of vertices in V' is connected by an edge
 (D) All pairs of vertices in V' are not connected by an edge

- Q37** What is the worst case running time of Insert and Extract-min, in an implementation of a priority queue using an unsorted array? Assume that all insertions can be accommodated.
 (A) $\theta(1)$, $\theta(n)$



- (B) $\theta(n)$, $\theta(n)$
 (C) $\theta(1)$, $\theta(1)$
 (D) $\theta(n)$, $\theta(n)$

Q38 In a B-Tree, each node represents a disk block. Suppose one block holds 8192 bytes. Each key uses 32 bytes. In a B-tree of order M there are M - 1 keys. Since each branch is on another disk block, we assume a branch is of 4 bytes. The total memory requirement for a non-leaf node is
 (A) $32M - 32$ (B) $36M - 32$
 (C) $36M - 36$ (D) $32M - 36$

Q39 Give asymptotic upper and lower bound for $T(n)$ given below. Assume $T(n)$ is constant for $n \leq 2$. $T(n) = 4T(\sqrt{n}) + \lg^2 n$
 (A) $T(n) = \theta(\lg(\lg^2 n) \lg n)$
 (B) $T(n) = \theta(\lg^2 n \lg n)$
 (C) $T(n) = \theta(\lg^2 n \lg \lg n)$
 (D) $T(n) = \theta(\lg(\lg n) \lg n)$

Q40 Consider a weighted directed graph. The current shortest distance from source S to node x is represented by $d[x]$. Let $d[v] = 29$, $d[u] = 15$, $w[u,v] = 12$. What is the updated value of $d[v]$ based on current information?
 (A) 29 (B) 27
 (C) 25 (D) 17

Q41 When using Dijkstra's algorithm to find shortest path in a graph, which of the following statement is not true?
 (A) It can find shortest path within the same graph data structure
 (B) Every time a new node is visited, we choose the node with smallest known distance/ cost (weight) to visit first
 (C) Shortest path always passes through least number of vertices
 (D) The graph needs to have a non-negative weight on every edge

Q42 The time complexity to multiply two polynomials of degree n using Fast Fourier transform method is :
 (A) $\theta(n \lg n)$

- (B) $\theta(n^2)$
 (C) $\theta(n)$
 (D) $\theta(\lg n)$

Q43 Consider the following grammars :

$$\begin{aligned} G_1 : S &\rightarrow aSb|bSa|aa \\ G_2 : S &\rightarrow aSb|bSa|SS | \lambda \\ G_3 : S &\rightarrow aSb|bSa|SS | a \\ G_4 : S &\rightarrow aSb|bSa|SS|SSS|\lambda \end{aligned}$$

Which of the following is correct w.r.t. the above grammars?

- (A) G_1 and G_3 are equivalent
 (B) G_2 and G_3 are equivalent
 (C) G_2 and G_4 are equivalent
 (D) G_3 and G_4 are equivalent

Q44 Consider the language $L = \{a^n b^{n-3} \mid n > 2\}$ on $\Sigma = \{a, b\}$. Which one of the following grammars generates the language L ?

- (A) $S \rightarrow aA|a, A \rightarrow aAb|b$
 (B) $S \rightarrow aaA|\lambda, A \rightarrow aAb|\lambda$
 (C) $S \rightarrow aaaA|a, A \rightarrow aAb|\lambda$
 (D) $S \rightarrow aaaA, A \rightarrow aAb | \lambda$

Q45 Consider the following grammar :

$$\begin{aligned} S &\rightarrow 0A | 0BB \\ A &\rightarrow 00A | \lambda \\ B &\rightarrow 1B | 11C \\ C &\rightarrow B \end{aligned}$$

Which language does this grammar generate?

- (A) $L((00)^* 0 + (11)^* 1)$
 (B) $L(0(11)^* + 1(00)^*)$
 (C) $L((00)^* 0)$
 (D) $L(0(11)^* 1)$

Q46 Consider $\Sigma = \{w, x\}$ and $T = \{x, y, z\}$. Define homomorphism h by :

$$\begin{aligned} h(x) &= xzy \\ h(w) &= zxyy \end{aligned}$$

If L is the regular language denoted by

$r = (w + x^*)(ww)^*$, then the regular language $h(L)$ is given by

- (A) $(zxyy + xzy)(zxyy)$
 (B) $(zxyy + (xzy)^*)(zxyyzxyy)^*$
 (C) $(zxyy + xzy)(zxyy)^*$



(D) $(xyxy + (xzy)^*)(xyxyzxyy)$

Q47 Consider the following statements with respect to the language $L = \{a^n b^n \mid n \geq 0\}$

$S_1 : L^2$ is context free language

$S_2 : L^k$ is context-free language for any given $k \geq 1$

$S_3 : \bar{L}$ and L^* are context free languages

Which one of the following is correct?

(A) only S_1 and S_2

(B) only S_1 and S_3

(C) only S_2 and S_3

(D) S_1, S_2 and S_3

Q48 Consider the following languages :

$L_1 = \{a^n b^n c^m\} \cup \{a^n b^m c^m\}, n, m \geq 0$

$L_2 = \{ww^R \mid w \in \{a, b\}^*\}$ Where R

represents reversible operation.

Which one of the following is (are) inherently ambiguous language(s)?

(A) only L_1

(B) only L_2

(C) both L_1 and L_2

(D) neither L_1 nor L_2

Q49 Let $G=(V, T, S, P)$ be any context-free grammar without any λ -productions or unit productions. Let K be the maximum number of symbols on the right of any production in P. The maximum number of production rules for any equivalent grammar in Chomsky normal form is given by: Where $1 - I$ denotes the cardinality of the set.

(A) $(K - 1)|P| + |T| - 1$

(B) $(K - 1)|P| + |T|$

(C) $K|P| + |T| - 1$

(D) $K|P| + |T|$

Q50 Consider the following language families :

$L_1 \equiv$ The context - free languages

$L_2 \equiv$ The context - sensitive languages

L_3

$\equiv$ The recursively enumerable languages

$L_4 \equiv$ The recursive languages

Which one of the following options is correct?

(A) $L_1 \subseteq L_2 \subseteq L_3 \subseteq L_4$

(B) $L_2 \subseteq L_1 \subseteq L_3 \subseteq L_4$

(C) $L_1 \subseteq L_2 \subseteq L_4 \subseteq L_3$

(D) $L_2 \subseteq L_1 \subseteq L_4 \subseteq L_3$

Q51 Consider the following statements :

S_1 : There exists no algorithm for deciding if any two Turing machines M_1 and M_2 accept the same language.

S_2 : Let M_1 and M_2 be arbitrary Turing machines. The problem to determine $\subseteq L(M_2)$ is undecidable.

Which of the statements is (are) correct?

(A) Only S_1

(B) Only S_2

(C) Both S_1 and S_2

(D) Neither S_1 nor S_2

Q52 Let $A = \{001, 0011, 11, 101\}$ and

$B = \{01, 111, 111, 010\}$. Similarly, let

$C = \{00, 001, 1000\}$ and $D = \{0, 11, 011\}$.

Which of the following pairs have a post-correspondence solution?

(A) Only pair (A, B)

(B) Only pair (C, D)

(C) Both (A, B) and (C, D)

(D) Neither (A, B) nor (C, D)

Q53 Which of the following class of IP address has the last address as 223.255 .255 .255 ?

(A) Class A

(B) Class B

(C) Class C

(D) Class D

Q54 Consider a subnet with 720 routers. If a three-level hierarchy is chosen, with eight clusters, each containing 9 regions of 10 routers, then total number of entries in hierarchical table of each router is

(A) 25

(B) 27

(C) 53

(D) 72

Q55 Piconet is a basic unit of a bluetooth system consisting of----- master node and up to ----- active slave nodes.

(A) one, five

(B) one, seven

(C) two, eight

(D) one, eight

Q56



A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

- (A) 1, 000, 000bps
- (B) 2,000,000bps
- (C) 12, 000. 000bps
- (D) 1, 200, 00, 000bps

Q57 The full form of ICANN is

- (A) Internet Corporation for Assigned Names and Numbers
- (B) Internet Corporation for Assigned Numbers and Names
- (C) Institute of Cooperation for Assigned Names and Numbers
- (D) Internet Connection for Assigned Names and Numbers

Q58 According to Dempster-Shafer theory for uncertainty management,

Where $\text{Bel}(A)$ denotes Belief of event A.

- (A) $\text{Bel}(A) + \text{Bel}(\neg A) \leq 1$
- (B) $\text{Bel}(A) + \text{Bel}(\neg A) \geq 1$
- (C) $\text{Bel}(A) + \text{Bel}(\neg A) = 1$
- (D) $\text{Bel}(A) + \text{Bel}(\neg A) = 0$

Q59 Consider the following statements:

$S_1 : \forall x P(x) \vee \forall x Q(x)$ and

$\forall x (P(x) \vee Q(x))$ are not logically equivalent.

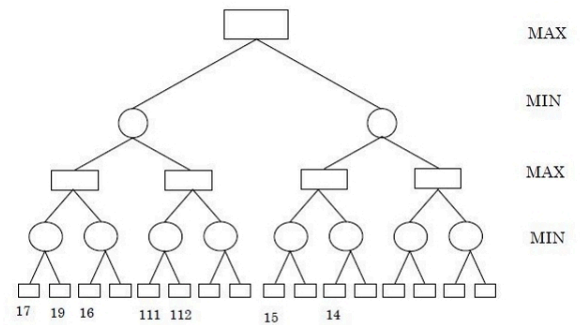
$S_2 : \exists x P(x) \wedge \exists x Q(x)$ and

$\exists x (P(x) \wedge Q(x))$ are not logically equivalent

Which of the following statements is/are correct?

- (A) Only S_1
- (B) Only S_2
- (C) Both S_1 and S_2
- (D) Neither S_1 nor S_2

Q60 Consider the game tree given below



Here $\circ$ and $\square$ represent MIN and MAX nodes respectively. The value of the root node of the game tree is:

- (A) 14
- (B) 17
- (C) 111
- (D) 112

Q61 Consider the following models:

M1: Mamdani model

M_2 : Takagi - Sugeno-Kang model

M3: Kosko's additive model (SAM)

Which of the following option contains examples of additive rule model?

- (A) Only M_1 and M_2
- (B) Only M_2 and M_3
- (C) Only M_1 and M_3
- (D) M_1, M_2 , and M_3

Q62 A fuzzy conjunction operators, $t(x, y)$, and a fuzzy disjunction operator, $s(x, y)$, form a pair if they satisfy:

$$t(x, y) = 1 - s(1 - x, 1 - y)$$

If $t(x, y) = \frac{xy}{(x+y-xy)}$ then $s(x, y)$ is given by

- (A) $\frac{x+y}{1-xy}$
- (B) $\frac{x+y-2xy}{1-xy}$
- (C) $\frac{x+y-xy}{1-xy}$
- (D) $\frac{x+y-xy}{1+xy}$

Q63 Let W_{ij} represents weight between node i at layer k and node j at layer $(k - 1)$ of a given multilayer perceptron. The weight updation using gradient descent method is given by

- (A) $W_{ij}(t+1) = W_{ij}(t) + \alpha \frac{\partial E}{\partial W_{ij}}, 0 \leq \alpha \leq 1$
- (B) $W_{ij}(t+1) = W_{ij}(t) - \alpha \frac{\partial E}{\partial W_{ij}}, 0 \leq \alpha \leq 1$
- (C) $W_{ij}(t+1) = \alpha \frac{\partial E}{\partial W_{ij}}, 0 \leq \alpha \leq 1$
- (D) $W_{ij}(t+1) = -\alpha \frac{\partial E}{\partial W_{ij}}, 0 \leq \alpha \leq 1$



Q64 The order of schema ?10?101? and ???0??1 are----- and -----respectively.

- (A) 5,3 (B) 5,2
(C) 7,5 (D) 8,7

Q65 Let the population of chromosomes in genetic algorithm is represented in terms of binary number. The strength of fitness of a chromosome in decimal form, x , is given by

$$Sf(x) = \frac{f(x)}{\sum f(x)} \text{ where } f(x) = x^2$$

The population is given by P where:

$$P = \{(01101, (11000), (01000), (10011)\}$$

The strength of fitness of chromosome (11000) is -----

- (A) 24 (B) 576
(C) 14.4 (D) 49.2

Q66 Consider the following statements :

S_1 : If a group $(G, *)$ is of order n , and $a \in G$ is such that $a^m = e$ for some integer $m \leq n$, then m must divide n .

S_2 : If a group $(G, *)$ is of even order, then there must be an element $a \in G$ such that $a \neq e$ and $a * a = e$.

Which of the statements is (are) correct?

- (A) Only S_1
(B) Only S_2
(C) Both S_1 and S_2
(D) Neither S_1 nor S_2

Q67 Consider the following statements with respect to duality in LPP :

- (a) The final simplex table giving optimal solution of the primal also contains optimal solution of its dual in itself.
(b) If either the primal or the dual problem has a finite optimal solution, then the other problem also has a finite optimal solution.
(c) If either problem has an unbounded optimum solution, then the other problem has no feasible solution at all.

Which of the statements is (are) correct?

- (A) only (a) and (b)

(B) only (a) and (c)

(C) only (b) and (c)

(D) (a), (b) and (c)

Q68 The Reduced Instruction Set Computer (RISC) characteristics are :

- (a) Single cycle instruction execution
(b) Variable length instruction formats
(c) Instructions that manipulates operands in memory
(d) Efficient instruction pipeline

Choose the correct characteristics from the options given below :

- (A) (a) and (b) (B) (b) and (c)
(C) (a) and (d) (D) (c) and (d)

Q69 Which of the following binary codes for decimal digits are self complementing?

- (a) 8421 code
(b) 2421 code
(c) excess-3 code
(d) excess-3 gray code

Choose the correct option :

- (A) (a) and (b) (B) (b) and (c)
(C) (c) and (d) (D) (d) and (a)

Q70 Consider the following statements with respect to approaches to fill areas on raster systems :

P: To determine the overlap intervals for scan lines that cross the area.

Q: To start from a given interior position and paint outward from this point until we encounter the specified boundary conditions.

Select the correct answer from the options given below :

- (A) P only
(B) Q only
(C) Both P and Q
(D) Neither P nor Q

Q71 Which of the following statements are true regarding C ++ ?

- (a) Overloading gives the capability to an existing operator to operate on other data types.
(b) Inheritance in object oriented programming provides support to reusability.



(c) When object of a derived class is defined, first the constructor of derived class is executed then constructor of a base class is executed.

(d) Overloading is a type of polymorphism.

Choose the correct option from those given below :

(A) (a) and (b) only

(B) (a), (b) and (c) only

(C) (a), (b) and (d) only

(D) (b), (c) and (d) only

Q72 Which of the following are legal statements in C programming language?

(a) $\text{int } *P = \&44;$

(b) $\text{int } *P = \&r;$

(c) $\text{int } P = \&a :$

(d) $\text{int } P = a;$

Choose the correct option :

(A) (a) and (b)

(B) (b) and (c)

(C) (b) and (d)

(D) (a) and (d)

Q73 Two concurrent executing transactions T_1 and T_2 are allowed to update same stock item say 'A' in an uncontrolled manner. In such scenario, following problems may occur :

(a) Dirty read problem

(b) Lost update problem

(c) Transaction failure

(d) Inconsistent database state

Which of the following option is correct if database system has no concurrency module and allows concurrent execution of above two transactions?

(A) (a), (b) and (c) only

(B) (c) and (d) only

(C) (a) and (b) only

(D) (a), (b) and (d) only

Q74 Identify the circumstances under which pre-emptive CPU scheduling is used :

(a) A process switches from Running state to Ready state

(b) A process switches from Waiting state to Ready state

(c) A process completes its execution

(d) A process switches from Ready to Waiting state

Choose the correct option :

(A) (a) and (b) only

(B) (a) and (d) only

(C) (c) and (d) only

(D) (a), (b), (c) only

Q75 The following multithreaded algorithm computes transpose of a matrix in parallel :

p Trans (X, Y, N)

if $N = 1$

then $Y[1,1] \leftarrow X[1,1]$

else partition X into four $(N/2) \times (N/2)$

submatrices $X_{11}, X_{12}, X_{21}, X_{22}$

partition Y into four $(N/2) \times (N/2)$ submatrices

$Y_{11}, Y_{12}, Y_{21}, Y_{22}$

spawn p Trans ($X_{11}, Y_{11}, N/2$)

spawn p Trans ($X_{12}, Y_{12}, N/2$)

spawn p Trans ($X_{21}, Y_{21}, N/2$)

spawn p Trans ($X_{22}, Y_{22}, N/2$)

What is the asymptotic parallelism of the algorithm?

(A) T_1/T_∞ or $\theta(N^2 / \lg N)$

(B) T_1/T_∞ or $\theta(N / \lg N)$

(C) T_1/T_∞ or $\theta(\lg N / N^2)$

(D) T_1/T_∞ or $\theta(\lg N / N)$

Q76 Consider the following statements :

(a) The running time of dynamic programming algorithm is always $\theta(\rho)$ where ρ is number of subproblems.

(b) When a recurrence relation has cyclic dependency, it is impossible to use that recurrence relation (unmodified) in a correct dynamic program.

(c) For a dynamic programming algorithm, computing all values in a bottom-up fashion is asymptotically faster than using recursion and memorization.

(d) If a problem X can be reduced to a known NP-hard problem, then X must be NP-hard.

Which of the statement(s) is (are) true?

(A) Only (b) and (a)

(B) Only (b)



- (C) Only (b) and (c)
(D) Only (b) and (d)

Q77 Consider the following statements :

- (a) Fiber optic cable is much lighter than copper cable.
(b) Fiber optic cable is not affected by power surges or electromagnetic interference.
(c) Optical transmission is inherently bidirectional.
Which of the statements is (are) correct?

- (A) Only (a) and (b)
(B) Only (a) and (c)
(C) Only (b) and (c)
(D) (a), (b) and (c)

Q78 Consider the following statements :

- (a) Windows Azure is a cloud-based operating system.
(b) Google App Engine is an integrated set of online services for consumers to communicate and share with others.
(c) Amazon Cloud Front is a web service for content delivery.
Which of the statements is (are) correct?

- (A) Only (a) and (b)
(B) Only (a) and (c)
(C) Only (b) and (c)
(D) (a), (b) and (c)

Q79 Consider the following statements with respect to network security :

- (a) Message confidentiality means that the sender and the receiver expect privacy.
(b) Message integrity means that the data must arrive at the receiver exactly as they were sent.
(c) Message authentication means the receiver is ensured that the message is coming from the intended sender.
Which of the statements is (are) correct?

- (A) Only (a) and (b)
(B) Only (a) and (c)
(C) Only (b) and (c)
(D) (a), (b) and (c)

Q80 Consider the following:

- (a) Trapping at local maxima

- (b) Reaching a plateau
(c) Traversal along the ridge.

Which of the following option represents shortcomings of the hill climbing algorithm?

- (A) (a) and (b) only
(B) (a) and (c) only
(C) (b) and (c) only
(D) (a), (b) and (c)

Q81 Consider the following learning algorithms :

- (a) Logistic regression
(b) Back propagation
(c) Linear regression

Which of the following option represents classification algorithms?

- (A) (a) and (b) only
(B) (a) and (c) only
(C) (b) and (c) only
(D) (a), (b) and (c)

Q82 Match List-I and List-II :

List I		List II	
(a)	Physical layer	(i)	Provide token management service
(b)	Transport layer	(ii)	Concerned with transmitting raw bits over a communication channel
(c)	Session layer	(iii)	Concerned with the syntax and semantics of the information transmitted
(d)	Presentation layer	(iv)	True end-to-end layer from source to destination

Choose the correct option from those given below :

- (A) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)



(B) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(C) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

(D) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

Q83 According to the ISO-9126 Standard Quality Model, match the attributes given in List-I with their definitions in List-II :

List I		List II	
(a)	Functionality	(i)	Relationship between layer of performance and amount of resources
(b)	Reliability	(ii)	Characteristics related with achievement of purpose
(c)	Efficiency	(iii)	Efforts needed to make for improvement
(d)	Maintainability	(iv)	Capability of software to maintain performance of software

Choose the correct option from the ones given below :

(A) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

(B) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

(C) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

(D) (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii)

Q84 Match the Agile Process models with the task performed during the model :

	List I		List II
(a)	Scrum	(i)	CRC crads
(b)	Adaptive software development	(ii)	Sprint backlog
(c)	Extreme programming	(iii)	the by/for/of/to> a(n)

(d)	Feature-driven development	(iv)	Time box release plan
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Choose the correct option from those given below :

(A) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

(B) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)

(C) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

(D) (a)-(i), (b)-(iv), (c)-(ii), (d)-(iii)

Q85 Match List-I with List-II :

	List I		List II
(a)	Frame attribute	(i)	to create link in HTML
(b)	<tr> tag	(ii)	for vertical alignment of content in cell
(c)	Valign attribute	(iii)	to enclose each row in table
(d)	<a> tag	(iv)	to specify the side of the table frame that dispaly border

Choose the correct option from those given below :

(A) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

(B) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

(C) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(D) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

Q86 An instruction is stored at location 500 with its address field at location 501. The address field has the value 400. A processor register R_1 contains the number 200. Match the addressing mode (List-I) given below with effective address (List-II) for the given instruction:

	List I		List II
(a)	Direct	(i)	200



(b)	Register indirect	(ii)	902
(c)	Indec with R ₁ as the index register	(iii)	400
(d)	Relative	(iv)	600

Choose the correct option from those given below :

- (A) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
 (B) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
 (C) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)
 (D) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

Q87 Match List-I and List-II :

	List I		List II
(a)	Isolated I/O	(i)	same set of control signal for I/O and memory communication
(b)	Memory mapped I/O	(ii)	separate instructions for I/O and memory communication
(c)	I/O interface	(iii)	require control signals to be transmitted between the communicating units
(d)	Asynchronous data transfer	(iv)	resolve the difference in central computer and peripherals

Choose the correct option from those given below :

- (A) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
 (B) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
 (C) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
 (D) (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii)

Q88 Match List-I with List-II :

	List I		List II
(a)	Micro operation	(i)	Specify micro operations
(b)	Micro programmed control unit	(ii)	Improve CPU utilization
(c)	Interrupts	(iii)	Control Memory
(d)	Micro instruction	(iv)	Elementary operation performed on data stored in register

Choose the correct option from those given below :

- (A) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
 (B) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
 (C) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
 (D) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

Q89

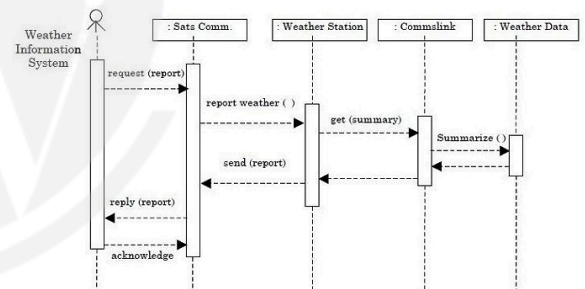


Figure 1

The sequence diagram given in Figure 1 for the Weather Information System takes place when an external system requests the summarized data from the weather station. The increasing order of lifeline for the objects in the system are:

- (A) Sat comms → Weather station → Commlink → Weather data
 (B) Sat comms → Comms link → Weather station → Weather data
 (C) Weather data → Comms link → Weather station → Sat Comms
 (D)

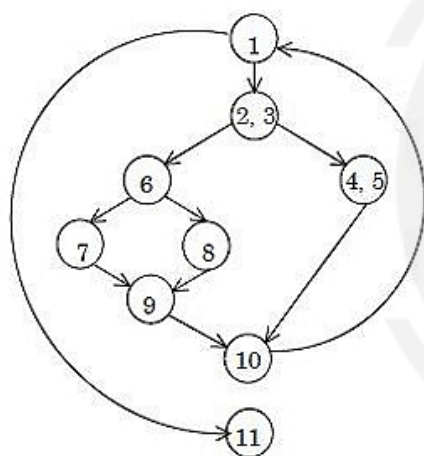


Weather data $\rightarrow$ Weather station $\rightarrow$ Comms link $\rightarrow$ Sat Comms

- Q90** The Data Encryption Standard (DES) has a function consists of four steps. Which of the following is correct order of these four steps?
- (A) an expansion permutation, S-boxes, an XOR operation, a straight permutation
 (B) an expansion permutation, an XOR operation, S-boxes, a straight permutation
 (C) a straight permutation, S-boxes, an XOR operation, an expansion permutation
 (D) a straight permutation, an XOR operation, S-boxes, an expansion permutation

Directions (91-95) Read the following passage and answer the given questions.

A flow graph F with entry node (1) and exit node (11) is shown below :



Flowgraph F

- Q91** How many nodes are there in flowgraph F?
 (A) 9 (B) 10
 (C) 11 (D) 12
- Q92** What is the cyclomatic complexity of flowgraph F?
 (A) 2 (B) 3
 (C) 4 (D) 5
- Q93** How many regions are there in flowgraph F?
 (A) 2 (B) 3
 (C) 4 (D) 5

- Q94** How many nodes are there in the longest independent path?
 (A) 6 (B) 7
 (C) 8 (D) 9
- Q95** How many predicate nodes are there and what are their names?
 (A) Three : (1, (2, 3), 6)
 (B) Three : (1, 4, 6)
 (C) Four : ((2, 3), 6, 10, 11)
 (D) Four : ((2, 3), 6, 9, 10)

Directions (96-100) Read the following passage and answer the given questions.

An organization needs to maintain database having five attributes A, B, C, D, E . These attributes are functionally dependent on each other for which functional dependency set F is given as :

$F : \{A \rightarrow BC, D \rightarrow E, BC \rightarrow D, A \rightarrow D\}$.
 Consider a universal relation $R(A, B, C, D, E)$ with functional dependency set F . Also all attributes are simple and take atomic values only.

- Q96** Minimal cover F' of functional dependency set F is
 (A) $F' = \{A \rightarrow B, A \rightarrow C, BC \rightarrow D, D \rightarrow E\}$
 (B) $F' = \{A \rightarrow BC, B \rightarrow D, D \rightarrow E\}$
 (C) $F' = \{A \rightarrow B, A \rightarrow C, A \rightarrow D, D \rightarrow E\}$
 (D) $F' = \{A \rightarrow B, A \rightarrow C, B \rightarrow D, C \rightarrow D, D \rightarrow E\}$
- Q97** Identify primary key of table R with functional dependency set F
 (A) BC (B) AD
 (C) A (D) AB
- Q98** Identify the normal form in which relation R belong to
 (A) 1 NF (B) 2 NF
 (C) 3 NF (D) BCNF
- Q99** Identify the redundant functional dependency in F



- (A) $BC \rightarrow D$
- (B) $D \rightarrow E$
- (C) $A \rightarrow D$
- (D) $A \rightarrow BC$

Q100 Assume that given table R is decomposed in two tables

$R_1(A, B, C)$ with functional dependency set

$f_1 = \{A \rightarrow B, A \rightarrow C\}$ and

$R_2(A, D, E)$ with FD set

$F_2 = \{A \rightarrow D, D \rightarrow E\}$.

Which of the following option is true w.r.t. given decomposition?

- (A) Dependency preservation property is followed
- (B) R_1 and R_2 are both in $2NF$
- (C) R_2 is in $2NF$ and R_3 is in $3NF$
- (D) R_1 is in $3NF$ and R_2 is in $2NF$



Answer Key

Q1 (B)
Q2 (C)
Q3 (B)
Q4 (A)
Q5 (C)
Q6 (B)
Q7 (A)
Q8 (B)
Q9 (D)
Q10 (C)
Q11 (A)
Q12 (D)
Q13 (D)
Q14 (A)
Q15 (D)
Q16 (B)
Q17 (B)
Q18 (B)
Q19 (C)
Q20 (B)
Q21 (D)
Q22 (B)
Q23 (C)
Q24 (A)
Q25 (D)
Q26 (B)
Q27 (B)
Q28 (B)
Q29 (C)
Q30 (B)
Q31 (D)

Q32 (B)
Q33 (C)
Q34 (B)
Q35 (A)
Q36 (C)
Q37 (A)
Q38 (B)
Q39 (C)
Q40 (B)
Q41 (C)
Q42 (A)
Q43 (C)
Q44 (D)
Q45 (C)
Q46 (B)
Q47 (D)
Q48 (A)
Q49 (B)
Q50 (C)
Q51 (C)
Q52 (A)
Q53 (C)
Q54 (A)
Q55 (B)
Q56 (B)
Q57 (A)
Q58 (C)
Q59 (C)
Q60 (B)
Q61 (B)
Q62 (B)



Q63 (B)
Q64 (B)
Q65 (D)
Q66 (C)
Q67 (D)
Q68 (C)
Q69 (B)
Q70 (C)
Q71 (C)
Q72 (C)
Q73 (D)
Q74 (A)
Q75 (A)
Q76 (B)
Q77 (A)
Q78 (B)
Q79 (D)
Q80 (D)
Q81 (A)

Q82 (C)
Q83 (C)
Q84 (A)
Q85 (C)
Q86 (A)
Q87 (C)
Q88 (A)
Q89 (C)
Q90 (B)
Q91 (A)
Q92 (C)
Q93 (C)
Q94 (C)
Q95 (A)
Q96 (A)
Q97 (C)
Q98 (B)
Q99 (C)
Q100 (D)



Hints & Solutions

Q1 Text Solution:

Transportation Problem:

A transportation problem is a special type of linear programming problem.

The goal is to minimize the transportation cost while satisfying supply and demand constraints.

Feasible Solution:

A feasible solution satisfies all supply and demand constraints.

A **basic feasible solution** is a feasible solution that corresponds to a set of independent allocations (non-zero variables in the solution matrix).

Number of Allocations in Basic Feasible Solution:

For a transportation problem with m rows (supplies) and n columns (demands), the **basic feasible solution**: $m+n-1$ allocations (non-zero values).

This is because the number of independent constraints is $m+n-1$, ensuring no degeneracy in the solution.

Non-Degeneracy:

A solution is **non-degenerate** if it has $m+n-1$ independent allocations.

If the number of allocations is fewer than $m+n-1$, the solution is **degenerate**.

Final Answer:

B. $m+n-1$, independent

Q2 Text Solution:

Option C: $x_1=1000, x_2=500, z=1500$

Substitute into **Constraint 1**:

$$x_1 + 2x_2 = 1000 + 2(500) = 1000 + 1000 = 2000 \text{ (satisfies } \leq 2000 \text{)}.$$

Substitute into **Constraint 2**:

$$x_1 + x_2 = 1000 + 500 = 1500 \text{ (satisfies } \leq 1500 \text{)}.$$

Substitute into **Constraint 3**:

$$x_2 = 500 \text{ (satisfies } \leq 600 \text{)}.$$

All constraints are satisfied.

Q3 Text Solution:

Option B is Correct

$$AB + A\bar{B} + \bar{A}C + AC$$

Simplifying the Equation

$$= A(B + \bar{B}) + \bar{A}C + AC$$

$$= A + \bar{A}C + AC$$

$$= A + AC + \bar{A}C$$

$$= A(1 + C) + \bar{A}C$$

$$= A + \bar{A}C$$

$$= A + C$$

The final expression does not depend on B. B is irrelevant because it was eliminated during the simplification process.

Q4 Text Solution:

Greatest Lower Bound (GLB):

For a set S , the GLB is the **largest element** in Z^+ that divides every element of S .

Least Upper Bound (LUB):

For a set S , the LUB is the **smallest element** in Z^+ that is divisible by every element of S .

Poset (Z^+ , divides):

The poset is based on divisibility,

$$a \leq b \Leftrightarrow a \text{ divides } b.$$

GLB and LUB for $A = \{3, 9, 12\}$

GLB of A:

The GLB is the **greatest common divisor (GCD)** of the elements in A .

$$\text{GCD}(3, 9, 12) = 3.$$

LUB of A:

The LUB is the **least common multiple (LCM)** of the elements in A .

$$\text{LCM}(3, 9, 12) = 36.$$

$$\text{GLB}(A) = 3, \text{LUB}(A) = 36.$$

GLB and LUB for $B = \{1, 2, 4, 5, 10\}$

GLB of B:

The GLB is the **GCD** of the elements in B .

$$\text{GCD}(1, 2, 4, 5, 10) = 1 \text{ (since 1 divides all elements)}.$$

LUB of B:

The LUB is the **LCM** of the elements in B .

$$\text{LCM}(1, 2, 4, 5, 10) = 20.$$

$$\text{GLB}(B) = 1, \text{LUB}(B) = 20.$$

Final Answer:

For A: $\text{GLB} = 3, \text{LUB} = 36$.

For B: $\text{GLB} = 1, \text{LUB} = 20$.



Q5 Text Solution:**1. Symmetry**

A relation R is symmetric if $(a,b) \in R \Rightarrow (b,a) \in R$.

- If a is a brother of b, it does not necessarily mean that b is a brother of a. For example, if b is a sister, then b is not a "brother."
- R is **not symmetric**.

2. Transitivity

A relation R is transitive if $(a,b) \in R$ and $(b,c) \in R \Rightarrow (a,c) \in R$.

- If a is a brother of b, and b is a brother of c, then a will also be a brother of c.
- R is **transitive**.

3. Equivalence Relation

A relation is an equivalence relation if it is reflexive, symmetric, and transitive.

- Reflexivity: $(a,a) \in R$ A person cannot be their own brother, so R is **not reflexive**.
- Symmetry: R is **not symmetric**.
- Transitivity: R is **transitive**.

Since R is not reflexive or symmetric, it is **not an equivalence relation**.

4. Partial Order Relation

A relation is a partial order if it is reflexive, antisymmetric, and transitive.

- Reflexivity: R is **not reflexive**
- Antisymmetry: R is antisymmetric if $(a,b) \in R$ and $(b,a) \in R \Rightarrow a=b$. R is **not antisymmetric**, as two distinct brothers a and b can satisfy $(a,b) \in R$ and $(b,a) \in R$.
- Transitivity: R is **transitive**.

Since R is not reflexive or antisymmetric, it is **not a partial order relation**.

Option C is correct.

Q6 Text Solution:

Option B is Correct

Using Kruskal's Algorithm,

List the Edge Set from the graph:

Edge Set = $\{2,3,4,5,6,7,8\}$

Sort the Edges by Weight

The edges are sorted in ascending order of their weights:

Sorted Edge Set = $\{2,3,4,5,6,7,8\}$

Add Edges to MST

Using Kruskal's Algorithm, add edges one by one while ensuring there are no cycles:

1. Add the edge with weight 2.
2. Add the edge with weight 3.
3. Add the edge with weight 4.
4. Add the edge with weight 6.

These edges connect all 5 vertices (since $V-1=4$ for 5 vertices).

Now, Calculate the Total Weight

Total Weight of MST = $2+3+4+6 = 15$

Q7 Text Solution:**Step 1: Key Properties of Trees**

A tree with V vertices has $V-1$ edges.

Degree Sum Property:

The sum of the degrees of all vertices in a tree is equal to $2 \times (\text{number of edges})$, or: $\sum \text{degrees} = 2E$, where E is the number of edges.

Step 2: Degrees of Vertices in the Tree

The tree has:

$2n$ vertices of degree 1,

$3n$ vertices of degree 2,

n vertices of degree 3.

Total Number of Vertices (V):

$$V = 2n + 3n + n = 6n$$

Total Degree of Vertices:

$$\sum \text{degrees} = (2n \cdot 1) + (3n \cdot 2) + (n \cdot 3) = 2n + 6n + 3n = 11n$$

Using the degree sum property:

$$\sum \text{degrees} = 2E.$$

$$11n = 2E \Rightarrow E = 11n/2$$

Step 3: Relationship Between Vertices and Edges

For a tree:

$$E = V - 1.$$

Substitute $V = 6n$:

$$E = 6n - 1$$

Step 4: Solve for n

From $E = 11n/2$ and $E = 6n - 1$, equate the two:

$$11n/2 = 6n - 1.$$

Multiply through by 2 to eliminate the fraction:



$$11n = 12n - 2.$$

$$n = 2$$

Step 5: Calculate V and E

Vertices (V):

$$V = 6n = 6 \times 2 = 12.$$

Edges (E):

$$E = V - 1 = 12 - 1 = 11.$$

Option A is correct.

Q8 Text Solution:

A **relation** on a set S is a subset of $S \times S$, which means it is a set of ordered pairs (a, b) where $a, b \in S$.

A **reflexive relation** requires that every element in S be related to itself. This means that for a relation R on S : $(a, a) \in R$ for all $a \in S$.

Counting the Reflexive Relations

For a set S with 4 elements ($S = \{a, b, c, d\}$):

The pairs $(a, a), (b, b), (c, c), (d, d)$ **must** be included in every reflexive relation.

For the remaining pairs in $S \times S$, we consider pairs of the form $(a, b), (a, c)$ etc., where $a \neq b$. Each such pair can either be:

Included in the relation, or

Excluded from the relation.

Total Number of Pairs in $S \times S$:

$|S \times S| = 4 \times 4 = 16$ (all possible pairs of ordered elements).

Reflexive Pairs:

There are 4 reflexive pairs:

$(a, a), (b, b), (c, c), (d, d)$

Remaining Pairs:

The remaining $16 - 4 = 12$ pairs can be independently included or excluded in the reflexive relation. For each of these 12 pairs, there are 2 choices (include or exclude).

Total Number of Reflexive Relations:

$$\text{Total Reflexive Relations} = 2^{12}$$

Q9 Text Solution:

Non-Pipelined System:

Time per task: 30ns

Time for 100 tasks: $T_{\text{non-pipelined}} = 100 \times 30 = 3000\text{ns}$.

Pipelined System:

- Pipeline has **4 segments**.
- Clock cycle: 10ns.

- Time to process the first task: $T_{\text{first task}} = 4 \times 10 = 40\text{ns}$.
- After the first task, a new task completes every clock cycle (10ns).

Total time for 100 tasks in a pipelined system:

$$T_{\text{pipelined}} = T_{\text{first task}} + (n-1) \times \text{clock cycle}.$$

For $n=100$:

$$T_{\text{pipelined}} = 40 + (100-1) \times 10 = 40 + 990 = 1030\text{ns}.$$

Step 3: Speedup Calculation

$$\text{Speedup} = \frac{T_{\text{non-pipelined}}}{T_{\text{pipelined}}}.$$

Substitute the values:

$$\text{Speedup} = \frac{3000}{1030} \approx 2.91.$$

Answer:

D. 2.91

Q10 Text Solution:

1. Addressing Mode Field:

- There are **2 addressing modes** (direct and indirect).
- To encode 2 options, we need:

$$\log_2(2) = 1 \text{ bit}.$$

- **Bits for Addressing Mode Field = 1.**

2. Operation Code (Opcode):

- Total size at 32 bits.

3. Register Code Part:

- The computer has **256 registers**.
- To specify one of 256 registers, we need:

$$\log_2(256) = 8 \text{ bits}.$$

- **Bits for Register Code Part = 8.**

4. Address Part:

- The memory has 2^{19} , so we need 19 bits to specify a memory address.
- **Bits for Address Part = 19.**

Remaining Bits for Opcode:



- Total instruction size is **32 bits**. Subtract the bits used for other fields:
- Opcode Bits=32–
(Addressing Mode+Register Code+Address Par
t)
- Opcode Bits=32–(1+8+19)=4bits.

Option C is correct.

Q11 Text Solution:

Fields in the Microinstruction:

1. Microoperation Field (F1 and F2):

- Each subfield F1 and F2 has **15 distinct microoperations**.
- To encode 15 microoperations, we need **4 bits** (since $2^4=16$, which can encode up to 15 operations).
- Total bits for F1 + F2 = 4+4 = 8 bits.

2. Condition Field (CD):

- The condition field is for **four status bits**.
- To select one out of 4 status bits, we need **2 bits** (since $2^2=4$).
- Total bits for CD=2 bits.

3. Branch Field (BR):

- The branch field specifies **4 options**.
- To encode 4 options, we need **2 bits** (since $2^2=4$).
- Total bits for BR=2 bits.

4. Address Field (AD):

- The address space has **128 memory words**.
- To address 128 words, we need **7 bits** (since $2^7 = 128$).
- Total bits for AD=7 bits.

Total Size of Microinstruction:

Add up the bits required for each field:

Total Size=F1 + F2 + CD + BR + AD

Total Size=4+4+2+2+7=19 bits

Option A is correct.

Q12 Text Solution:

Step 1: Convert the right-hand side to decimal

$(75)_8$ is in base 8. To convert it to decimal:

$$(75)_8 = 7 \times 8^1 + 5 \times 8^0 = 56 + 5 = 61$$

Thus, the equation becomes:

$$(142)_b + (112)_{b-2} = 61$$

Step 2: Convert $(142)_b$ to decimal

The value of $(142)_b$ in decimal is:

$$1 \cdot b^2 + 4 \cdot b^1 + 2 \cdot b^0 = b^2 + 4b + 2$$

Step 3: Convert $(112)_{b-2}$ to decimal

The value of $(112)_{b-2}$ in decimal is:

$$1 \cdot (b-2)^2 + 1 \cdot (b-2)^1 + 2 \cdot (b-2)^0$$

Simplify:

$$1 \cdot (b-2)^2 + 1 \cdot (b-2) + 2 = (b-2)^2 + (b-2) + 2$$

Expand $(b-2)^2$:

$$(b-2)^2 = b^2 - 4b + 4$$

$$(b-2)^2 + (b-2) + 2 = b^2 - 4b + 4 + b - 2 + 2 = b^2 - 3b + 4$$

Step 4: Write the equation in terms of b

Substituting the decimal values:

$$(b^2 + 4b + 2) + (b^2 - 3b + 4) = 61$$

$$b^2 + 4b + 2 + b^2 - 3b + 4 = 61$$

$$2b^2 + b + 6 = 61$$

Step 5: Solve for b

$$2b^2 + b + 6 = 61$$

$$2b^2 + b - 55 = 0$$

This is a quadratic equation.

$$b = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$$

Here, $A = 2$, $B = 1$, $C = -55$. Substituting:



$$b = \frac{-1 \pm \sqrt{1^2 - 4(2)(-55)}}{2(2)}$$

$$b = \frac{-1 \pm \sqrt{1 + 440}}{4}$$

$$b = \frac{-1 \pm \sqrt{441}}{4}$$

$$b = \frac{-1 \pm 21}{4}$$

Step 6: Find valid b

$$1. b = \frac{-1+21}{4} = \frac{20}{4} = 5$$

$$2. b = \frac{-1-21}{4} = \frac{-22}{4} = -5.5 (\text{not valid since } b \text{ must be positive})$$

Thus, b=5.

Option D is correct.

Q13 Text Solution:

Option D is Correct

Instruction at 010: CLA (Clear Accumulator)

The CLA instruction clears the accumulator register. So, Accumulator after execution will be 0000.

Instruction at 011: ADD 016

Add the contents of memory location 016 to the accumulator. Memory at 016 contains the value C1A5 (hexadecimal). Now, the Accumulator after execution will be 0000+C1A5.

Instruction at 012: BUN 014

Branch unconditionally to location 014. So, the Execution continues from location 014.

Instruction at 014: AND 017

Perform a bitwise AND operation between the accumulator and the contents of memory location 017. The Memory at 017 contains the value 93C6 (hexadecimal). Now, Perform bitwise AND: C1A5(Accumulator)&93C6(Memory at 017) = 8184

So, the Accumulator after execution: 8184.

Instruction at 015: BUN 013

Branch unconditionally to location 013.

Instruction at 013: HLT

Halt the execution.

The accumulator contains 8184 (hexadecimal) after the program halts.

Q14 Text Solution:

Option A is Correct

Let's Understand the Code First:

There are 3 Variables:

- i and j: Loop control variables.
- x: Initialized to 0 and modified in the nested loop.

There are 2 Loops, the Outer Loop Runs from i = 0 to i = 4 (i < 5) and Inner Loop Runs for j = 0 to j < i (dependent on i). The break inside the inner loop terminates the loop after the first iteration of the inner loop, i.e., only one iteration for j = 0 is executed.

Inside the inner loop, the expression $x += (i + j - 1)$ is executed. Since j = 0 (inner loop breaks after one iteration), the effective expression is: $x += (i + 0 - 1) = x += (i - 1)$

Now, Going with Execution Trace:

1. i = 0:

- Inner loop does not execute because $j < i$ (0 < 0) is false.
- x remains 0.

2. i = 1:

- Inner loop runs for j = 0.
- $x += (1 - 1) = x + 0 = 0$.
- Breaks after the first iteration.

3. i = 2:

- Inner loop runs for j = 0.
- $x += (2 - 1) = x + 1 = 1$.

4. i = 3:

- Inner loop runs for j = 0.
- $x += (3 - 1) = x + 2 = 3$.

5. i = 4:

- Inner loop runs for j = 0.



- $x = -(4-1) = x+3 = 6$.

After all iterations, $x=6$.

Q15 Text Solution:

D. Object of derived class B can access public base class data

- False: Because with **protected inheritance**, the public members of the base class A are inherited as **protected** in the derived class B. Therefore, they are not accessible to objects of B directly.

Q16 Text Solution:

<script>

- The <script> tag is used to embed JavaScript code in an HTML document.
- It is designed specifically for executing JavaScript, either inline or from an external file.

Q17 Text Solution:

Step 1: Assign Region Codes

The Cohen-Sutherland algorithm assigns **region codes** to determine the position of each endpoint relative to the clipping window.

Region Code Format:

Region code = TOP BOTTOM RIGHT LEFT (4 bits)

1 indicates that the point is outside in that direction.

0 indicates that the point is inside in that direction.

Calculate Region Codes:

Point $(-1, 0)$:

$x=-1$, which is **left** of the window ($x<0$).

- $y=0$, which is within the vertical bounds ($0 \leq y \leq 3$).
- Region code = 001

Point $(4,5)$:

$x=4$, which is within the horizontal bounds ($0 \leq x \leq 5$).

- $y=5$, which is **above** the window ($y>3$).
- Region code = 1000

Step 2: Apply Clipping

The line segment intersects the clipping boundaries.

Intersection with $x=0$ (Left Edge):

Using the line equation, we calculate the y-coordinate for $x=0$:

$$\text{Line Equation: } y - y_1 = m(x - x_1)$$

$$x_1=-1, y_1=0, x_2=4, y_2=5$$

$$\text{Slope, } m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{5 - 0}{4 - (-1)} = 1$$

$$\text{At } x=0: y = 0 + 1(0 - (-1)) = 1$$

Intersection point: $(0,1)$

Intersection with $y=3$ (Top Edge):

Using the line equation, we calculate the x-coordinate for $y=3$:

$$x = x_1 + \frac{1}{m}(y - y_1)$$

At $y=3$:

$$x = -1 + \frac{1}{1}(3 - 0) = -1 + 3 = 2$$

Intersection point: $(2,3)$

Step 3: Final Clipped Line

The clipped line segment connects the points: $(0,1)$ and $(2,3)$

Option B is correct.

Q18 Text Solution:

Sutherland-Hodgman Algorithm:

- This algorithm is used for **polygon clipping**, not line clipping.
- It works by clipping a polygon edge by edge against the clipping window.

Q19 Text Solution:

Aspect Ratio Formula: The aspect ratio of an image is the ratio of its width to its height. For resizing,

$$\text{New Height} = \text{New Width} \times \frac{\text{Original Height}}{\text{Original Width}}$$

Given:



Original Width = 1024 pixels

Original Height = 768 pixels

New Width = 640 pixels

Calculate New Height: Using the formula:

$$\text{New Height} = 640 \times \frac{768}{1024}$$

New Height = $640 \times 0.75 = 480$ pixels

Option C is correct.

Q20 Text Solution:

Query Optimizer (Correct Answer):

Rearranging the query to minimize cost (e.g., time, space).

Choosing efficient execution plans based on available indexes, algorithms, and cost estimations.

Eliminating redundancies in the query and ensuring an optimal way to execute it.

Q21 Text Solution:

Explanation:

$R1(x, y) = 50$ tuples and $R2(y, z) = 30$ tuples

Formula:

Maximum number of tuples = Number of tuples in

$R1 \times \text{Number of tuples in } R2 = 50 \times 30 = 1500$.

The Final Answer is: 1500

Q22 Text Solution:

Explanation:

Using Natural Join, Find all entries with Department no D1 and D2. After using a Group of these tuples with DeptNo. Select all these tuples having a count > 5.
 $(\text{*count} > 5) \rightarrow (\text{* for all})$ and (>5) indicates greater than 5).

The Final Answer is: B

Q23 Text Solution:

Explanation:

Table R1 to R2 using One to Many Relationship using 2 Table

In Table R1, Multivalued attribute in B using 1 Table

Total Table required = $2+1 = 3$ Table

The Final Answer is :3

Q24 Text Solution:

Explanation:

The semaphore starts with a value of 8.

Wait() Operations $\rightarrow$ Each wait() operation reduces the value by 1.

Signal() Operations $\rightarrow$ Each signal() operation increases the value by 1.

According to the question:

After 3 wait() Operations $\rightarrow$ Value +

$(-1) \cdot \text{Operations} = 8 + (-1)(3) = 8 + (-3) = 8 - 3 = 5$.

Now semaphore starts with a value of 5.

After 4 signal() Operations $\rightarrow$ Value +

$(+1) \cdot \text{Operations} = 5 + (1)(4) = 5 + (+4) = 5 + 4 = 9$.

The Final Answer is: 9

Q25 Text Solution:

Explanation:

User details are not necessary for the encryption process itself; they may be used for authentication or access control but are not required for the encryption algorithm.

Encryption algorithm used in KEY, Message, and Ciphertext.

KEY is Used to encrypt and decrypt data.

The message is the actual data that needs to be encrypted.

Ciphertext is the result of encrypting the message.

The Final Answer is: User details

Q26 Text Solution:

Explanation:

Linux supports preemptive priority scheduling and time-sharing CPU scheduling.

It uses a combination of both techniques to manage process scheduling efficiently.

The Final Answer is: Preemptive priority scheduling and time-sharing CPU scheduling

Q27 Text Solution:

Explanation:

For TLB hits:

Contribution to Effective memory access time (EMAT) from TLB hits = $80\% = 0.8$

Hit Access time = TLB Access time + Memory Reference Time



→ $20 + 80 = 100$ Nanoseconds

Now, TLB hits = $\text{EMAT}(\text{TLB hits}) \times \text{Hit Access time} = 0.8 \times 100 = 80$ nanoseconds

For TLB misses:

Contribution to Effective memory access

time(EMAT) from TLB misses = $20\% = 0.2$

Miss Access time = TLB Access time + $2 \times$ Memory Reference Time

→ $20 + 2 \times 80 = 20 + 160 = 180$ Nanoseconds

Now, TLB misses = $\text{EMAT}(\text{TLB misses}) \times \text{Hit Access time} = 0.2 \times 180 = 36$ nanoseconds

Total EMAT(Effective memory access time) = $80 + 36 = 116$ Nanoseconds

The Final Answer is: B

Q28 Text Solution:

Explanation:

| | Maximum Needs | Current Needs |
|-------|---------------|---------------|
| p_0 | 10 | 5 |
| p_1 | 4 | 2 |
| p_2 | 9 | 2 |

Total Resources: 12 Magnetic tapes

Allocated Resources: Sum of all current Needs = $(5+2+2) = 9$.

Formula:

Available Resources: Total Resources – Allocated Resources

Available Resources: $12 - 9 = 3$

Step 1: Calculate the Remaining Needs for each Process:

| | | | |
|------------|----------------------------|------------|---|
| Need p_0 | Max p_0 – Allocate p_0 | $= 10 - 5$ | 5 |
| Need p_1 | Max p_1 – Allocate p_1 | $= 4 - 2$ | 2 |
| Need p_2 | Max p_2 – Allocate p_2 | $= 9 - 2$ | 7 |

Step 2: Safe sequence determination:

Start with Available = 3

Select a process where the remaining need is less than or equal to the available resource.

Step 3

First Allocation:

p_1 can execute because its need(2) $\leq$ Available (3) is True, After p_1 finishes, it releases its resources, adding 2 to Available is $3+2$ equals 5.

Second Allocation:

p_0 can execute because its need(5) $\leq$ Available (5) is True, After p_0 finishes, it releases its resources, adding 5 to Available is $5+5$ equals 10.

Third Allocation:

p_2 can execute because its need(7) $\leq$ Available (10) is True, After p_2 finishes, it releases its resources, adding 2 to Available is $10+2$ equals 12.

Safe Sequence: p_1, p_0, p_2

The Final Answer is: Option B

Q29 Text Solution:

Explanation:

Cooperative processes often use shared memory and message passing for efficient and structured communication.

Interprocess communication (IPC) uses shared memory or message passing.

Shared memory allows processes to exchange data by accessing a common memory region, offering speed but needing synchronization to avoid conflicts.

Message passing involves sending and receiving messages, making it easier for distributed systems.

The Final Answer is: C

Q30 Text Solution:

Explanation:

Round Robin time slice = 2 ms

Turn Around Time = Completion time - Arrival time

Waiting Time = Turnaround Time – Burst Time

| Process | Arrival Time | Burst Time | Completion Time (CT) | Turn Around Time (TAT) | Waiting Time |
|---------|--------------|---------------------------------|----------------------|------------------------|--------------|
| p_1 | 0 | $3 \rightarrow 1 \rightarrow 0$ | 5 | $5 - 0 = 5$ | $5 - 3 = 2$ |



| | | | | | |
|----|---|---------------|----|-----------|-----------|
| P2 | 2 | 4 → 2 → 0 | 7 | 7-2 = 5 | 5 - 4 = 1 |
| P3 | 5 | 5 → 3 → 1 → 0 | 12 | 12- 5 = 7 | 7- 5 = 2 |
| | | Total = 12 | | | |

Gantt Chart:

| | | | | | | |
|----|----|----|----|----|----|----------|
| P1 | P2 | P1 | P2 | P3 | P3 | P3 |
| 0 | 2 | 4 | 5 | 7 | 9 | 11
12 |

$$\text{Average Turn around time} = \frac{5 + 5 + 7}{3} = 5.66$$

$$\text{Average waiting time} = \frac{2 + 1 + 2}{3} = 1.66$$

The Final Answer is: 5.66, 1.66

Q31 Text Solution:

Explanation:

When a program makes a system call, it needs a way to send information to the operating system.

These are used to pass the parameters as below:

1. Using Register - It is stored directly in the CPU Register, It is very fast but can only handle a limited number of parameters due to the limited number of registers.

2. Using a Block in Memory: It is stored in a designated block of memory, and the address of this block is passed to the Operating System through a register.

3. Using Stack: It is pushed onto the program's stack, and the Operating System retrieves them when processing the system call.

i.e., we can say that the method used to pass any number of parameters to the operating system through system calls is by using a block in memory and the stack.

The Final Answer is: Block in main memory and stack

Q32 Text Solution:

Explanation:

The Java Virtual Machine needs two key components to execute Java code are below:

A. Class Loader → It loads the Java bytecode (compiled .class files) into memory from the

storage.

B. Java Interpreter → It interprets the bytecode and converts it into machine code that the computer can execute.

The Final Answer is: Class loader and Java Interpreter

Q33 Text Solution:

Explanation:

In the given restaurant order scenario, a sequence will typically have the following objects involved:

1. Customer: The person who interacts with the system by reading the menu, placing the order, requesting the bill, etc.
2. Menu: While not typically considered a separate object in the diagram, it may be implied as part of the customer interaction.
3. Order System: Handles the order placement and communication between the customer and the kitchen.
4. Kitchen: The part of the system responsible for preparing the ordered items.
5. Bill: Represents the bill generation and payment process.

The Final Answer is: 5

Q34 Text Solution:

Explanation:

An **Activity** chart is a project schedule representation that presents a project plan as a directed graph. The critical path is **Longest** the sequence of **Dependent** tasks and it defines project **Duration**.

The Final Answer is: Activity, Longest, Dependent, Duration

Q35 Text Solution:

Explanation:

$$\Rightarrow a^b \pmod{n} = 7^{17} \pmod{561} \text{ using value } a=7, b=17, \text{ and } n=561$$

$$\text{Base case: } 7 \pmod{561} = 7$$

$$\text{Square it: } 7^2 \pmod{561} = (7 \times 7) \pmod{561} = 49 \pmod{561} = 49$$

$$\text{Square again: } 7^4 \pmod{561} = (7^2 \times 7^2) \pmod{561} = 49 \times 49 \pmod{561} = 2401 \pmod{561} = 157$$



Square again: $7^8 \bmod 561 = (7^4 \times 7^4) \bmod 561 = 157 \times 157 \bmod 561 = 24649 \bmod 561 = 526$
 Square again: $7^{16} \bmod 561 = (7^8 \times 7^8) \bmod 561 = 526 \times 526 \bmod 561 = 276676 \bmod 561 = 103$
 Now, Substitute value $7^{16} \bmod 561 = 103$
 Calculate: $7^{17} \bmod 561 = 7^{16} \times 7 \bmod 561 = 103 \times 7 \bmod 561 = 721 \bmod 561 = 160$.

The Final Answer is: Option A

Q36 Text Solution:

Explanation:

A clique in an undirected graph is a subset of vertices V' such that every pair of distinct vertices in V' is adjacent, i.e. Each pair of vertices is connected by an edge.

The Final Answer is: C

Q37 Text Solution:

Explanation:

Insert operation: In an unsorted array, inserting an element takes constant time, $\theta(1)$ because we can simply add the new element at the end of the array.

Extract-min operation: To extract the minimum, we must search through the entire array to find the minimum value, which takes linear time, $\theta(n)$.

The Final Answer is: $\theta(1), \theta(n)$

Q38 Text Solution:

Explanation:

B- Tree of order M :

Each key uses 32 bytes, and there are M-1 Keys (non-leaf Node) = $32(M-1) = 32M-32$

Each child required 4 bytes, For M branches, the total memory required for branches is $M \times 4 = 4M$.

Total Memory = $(32M-32) + 4M = 36M-32$

therefore, The total Memory requirement for a non-leaf node is $36M-32$ bytes.

The Final Answer is: B

Q39 Text Solution:

Explanation:

Let $m = \log n$ so that $n = 2^m$, then we have

$$\sqrt{n} = \sqrt{2^m} = (2^m)^{\frac{1}{2}} = 2^{\frac{m}{2}}.$$

$$T(n) = 4T(\sqrt{n}) + \lg^2 n \dots\dots (1)$$

Substitute equation(1), $m = \log n$, $n = 2^m$, and

$$\sqrt{n} = \sqrt{2^m} = (2^m)^{\frac{1}{2}} = 2^{\frac{m}{2}}$$

$$\Rightarrow T(2^m) = 4 T(2^{m/2}) + (m)^2 \dots\dots (2)$$

Master Theorem Recurrences:

$$T(m) = a. T\left(\frac{m}{b}\right) + f(m)$$

Case 1: If $f(m) = O(m^{\log_b a - \epsilon})$ for $\epsilon > 0$ then

$$T(n) = \theta(m^{\log_b a})$$

Case 2: If $f(m) = \Theta(m^{\log_b a} \cdot \log_n k)$ for some k ; such that

$$(A) k \geq 0; \text{ then } T(n) = \Theta(m^{\log_b a} \log m)$$

Generate new function, $S(m) = T(2^m)$, Substitute equation (2)

$$\Rightarrow S(m) = 4 S\left(\frac{m}{2}\right) + (m)^2$$

Comparing on Master Theorem Recurrences : $a =$

$$4, b=2, \text{ and } f(m) = (m)^2$$

$$\text{Now calculate } \log_b a = \log_2 4 = 2.$$

Now, compare $f(m) = m^2$ with $m^{\log_b a} = m^2$

$f(m)$ is exactly m^2 , which matches $m^{\log_b a}$

Thus, the second case of the master theorem applies, which gives us: $S(m) = \Theta(m^2 \log m)$

Convert Back to n : Since $m = \log n$, we have $S(m) = T(n)$, therefore:

Thus, the asymptotic upper and lower bound for $T(n) = \Theta((\lg n)^2 \lg(\lg n))$.

The Final Answer is: Option C

Q40 Text Solution:

Explanation:

According to Question Information:

$d[v] = 29$ (Current shortest distance from source to node v)

$d[u] = 15$ (Current shortest distance from source to node u)

$w[u,v] = 12$ (weight of edges from u to v)

Formula: $d[v] = \min(d[v], d[u] + w(u,v))$

$$d[v] \Rightarrow \min(29, 15+12) = \min(29, 27) = 27$$

The updated value of $d[v]$ based on current information is 27.

The Final Answer is: 27

Q41 Text Solution:

Option- C (Shortest path always passes through the least number of vertices)

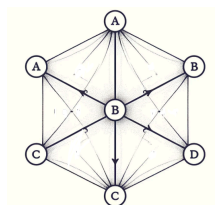
Explanation: The shortest path in a graph minimizes the **total weight (cost)** of the edges along the path rather than the **number of vertices**. A path with fewer vertices may have a higher total weight than another one with more



vertices.

Example : Consider a graph with vertices: A, B, C, D

Edges (weighted): From A to B= 1, From B to C= 1, From C to A= 5



Problem: Determine the shortest path from A to C.

1. Path 1: $A \rightarrow B \rightarrow C$.
 - Total weight = $1+1= 2$
 - Number of vertices = 3.
2. Path 2: $A \rightarrow C$
 - Total weight = 5
 - Number of vertices = 2.

Analysis: - Path 1 has more vertices but lower overall weight (2).

- Path 2 contains fewer vertices but a greater total weight (5).

Q42 Text Solution:

Option- $A(\theta(n \lg n))$

Explanation: The steps below describe how to multiply two polynomials of degree n using the Fast Fourier Transform (FFT).

1. **Transforming the polynomials** to the frequency domain by using the FFT- This step's time complexity is: $\theta(n \log n)$
2. **Applying point-wise multiplication** to the adjusted coefficients- This step involves multiplying 'n' coefficients, which takes $\theta(n)$ time.
3. **Use the inverse FFT** to convert the output to the time domain. Like the FFT, this step requires $\theta(n \log n)$. The FFT and inverse FFT are the most time-consuming operations, each taking $\theta(n \log n)$

Q43 Text Solution:

Option- **C(G_2 and G_4 are equivalent)**

Explanation: G_2 and G_4 produce the same language, making them equivalent. Both

grammars generate the same set of strings: all strings with equal numbers of a 's and b 's, including λ .

To justify why C: **G_2 and G_4 are equivalent** let's analyze the languages generated by these grammars in more detail:

Grammar G_2 :

$$S \rightarrow aSb \mid bSa \mid SS \mid \lambda$$

Base case: λ (the empty string) is part of the language.

Recursive rules:

$S \rightarrow aSb$: Generates strings where a and b are paired symmetrically.

$S \rightarrow SS$: Allows concatenation of any two valid substrings generated by S .

Language Generated: All strings with equal numbers of a 's and b 's in any order, including the empty string.

Examples: $\lambda, ab, ba, abab, aabbab$, etc.

Grammar G_4 :

$$S \rightarrow aSb \mid bSa \mid SS \mid SSS \mid \lambda$$

Base case: λ is included, just like in G_2 .

Recursive rules:

$S \rightarrow aSb$: Generates symmetric strings with a and b paired.

$S \rightarrow SS$: Allows concatenation of two substrings (same as G_2)

$S \rightarrow SSS$: Further generalizes concatenation by allowing three substrings.

Language Generated: This is an expansion of G_2 , but it doesn't change the language because SS already allows any concatenation. Adding SSS is redundant for generating all strings with equal numbers of a 's and b 's.

Examples: Same as G_2

Q44 Text Solution:

Option- **D($S \rightarrow aaaA, A \rightarrow aAb \mid \lambda$)**

Explanation:

$S \rightarrow aaaA, A \rightarrow aAb \mid \lambda$

The grammar ensures that 'S' always begins with



exactly three a's, as required by the condition $n > 2$.

The rule $A \rightarrow aAb$ ensures that for each additional "a", a corresponding "b" is generated, maintaining the difference of 3.

The rule $A \rightarrow \lambda$ correctly terminates the recursion.

This aligns perfectly with

$L = \{a^n b^{n-3} \mid n > 2\}$, making **Option D** the correct choice.

Q45 Text Solution:

Option- **C**($L((00)^*0)$)

Explanation: The key part of the grammar is $S \rightarrow 0A$. Here, A generates an even number of 0's (as it produces 00's repeatedly). So, starting with 0, followed by any even number of 00's, produces strings like:

0, 000, 0000, 000000

And so on...

These strings match the regular expression $L((00)^*0)$, which generates a string of 0 followed by any number of 00 pairs, ending with a final 0.

Thus, the language generated by the grammar corresponds to $L((00)^*0)$

Q46 Text Solution:

Option: **B**

$((xzyy + (xzy)^*)(xzyyzyxy)^*)$

Explanation:

1. **First part:** $(xzyy + (xzy)^*)$

$xzyy$: This represents $h(w)$, where w is replaced by $xzyy$.

$(xzy)^*$: This represents $h(x)^*$, where each x is replaced by xzy , and any number of x 's (including none) becomes repetitions of xzy .

2. **Second part:** $(xzyyzyxy)^*$:

This represents $h((ww)^*)$, where each w is replaced by $xzyy$. So, " ww " becomes $xzyyzyxy$, and the entire sequence can repeat zero or more times.

The language generated by the homomorphism consists of strings that either start with $xzyy$

(from w) or repetitions of xzy (from x^*), followed by any number of repetitions of $xzyyzyxy$ (from $(ww)^*$)

Q47 Text Solution:

Option- **D**(S_1, S_2 and S_3)

Explanation: Language $L = \{a^n b^n \mid n \geq 0\}$

This is a context-free language (CFL)

Statement S1: (L^2) refers to the concatenation of (L) with itself, i.e., $(L^2 = \{a^n b^n a^m b^m \mid n, m \geq 0\})$

- Context-free languages are closed to concatenation. As (L) is context-free, (L^2) , which is the concatenation of two (L) 's, will also be a context-free language.

Statement S2: - (L^k) indicates a concatenation of (L) with itself (k) times. Context-free languages are closed to finite concatenation, hence (L^k) for every $(k \geq 1)$ is also context-free.

Statement S3: L' and (L^*) are context-free languages. The counterpart to (L) , (L') , is equally context-free. In this situation, L is a deterministic context-free language (DCFL). DCFLs are closed under complement, making (L') context-free.

- (L^*) : The Kleene star operation on a context-free language (L) is always context-free due to its closure under the Kleene star.

Q48 Text Solution:

Option- **A**(Only L_1)

Explanation:

L_1 : There is an overlap between $\{a^n b^n c^m\}$ and $\{a^n b^m c^m\}$. Due to this overlap, it is impossible to construct an unambiguous context-free grammar L_1 , making it **inherently ambiguous**.

Q49 Text Solution:

Option- **B**($(K-1)|P|+|T|$)

Explanation:

1. **Splitting Long Productions:** A production containing (K) symbols on the right is divided into $(K - 1)$ binary productions. If there are $(|P|)$ productions, the total extra rules required are $((K - 1)|P|)$.

2. **Handling Terminals:** Each terminal in (T) may



require one extra production to ensure they appear alone (e.g., (A to a). This includes the ($|T|$) rules.

Total = split rules + terminal rules = $((K - 1)|P| + |T|)$

Q50 Text Solution:

Option- C($L_1 \subseteq L_2 \subseteq L_4 \subseteq L_3$)

Explanation: Analyzing the properties of the given language classes:

L_1 : Context-Free Languages (CFLs): A subset of context-sensitive languages that are defined by context-free grammars and recognizable by pushdown automata.

L_2 : Context-Sensitive Languages (CSLs): A superset of CFLs and a subset of recursively enumerable languages and is recognizable by linear bounded automata (LBA).

L_3 : Recursively Enumerable Languages (RE): Recognizable by Turing machines (TM) and includes all languages that TM can recognize (but may not halt).

L_4 : Recursive Languages: A proper subset of RE languages and is recognized by a TM that halts all inputs.

Q51 Text Solution:

Option- C(Both S_1 and S_2)

Explanation:

S_1 : The equivalency problem for Turing machines is undecidable as there is no technique to check if $L(M_1) = L(M_2)$. This is because determining the behavior of arbitrary Turing machines requires solving undecidable problems such as the halting problem. Therefore, S_1 is right.

S_2 : The subset issue for Turing machines $L(M_1) \subseteq L(M_2)$ is also undecidable. To determine $L(M_1) \subseteq L(M_2)$, one would need to verify that every string approved by M_1 is also accepted by M_2 , which cannot be algorithmically checked for all

circumstances. Therefore, S_2 is right.

Q52 Text Solution:

Option- A(Only pair (A,B))

Explanation: Solve the problem step by step by using indices $a_1, a_2, \dots$ for the elements in A and $b_1, b_2, \dots$ for the elements in B , and see if a **post-correspondence solution (PCS)** exists for any of the pairs.

Pair (A, B):

$A = \{a_1 = 001, a_2 = 0011, a_3 = 11, a_4 = 101\}$

$B = \{b_1 = 01, b_2 = 111, b_3 = 111, b_4 = 010\}$

We are looking for a sequence of indices $(a_1, a_2, \dots)$ and $(b_1, b_2, \dots)$ such that the concatenated strings from A and B are equal. Here, $a_3a_4 = b_2b_1 = 11101$ or $a_3a_4 = b_3b_1 = 11101$. Both sequences are equal, so we have found a post-correspondence solution (A, B) .

Q53 Text Solution:

Option- C(Class C)

Explanation:

For Class C:

Range: 192.0.0.0 to 223.255.255.255

So the address 223.255.255.255 falls within the range to, which is the **Class C range**.

Q54 Text Solution:

Option- A(25)

Explanation: Three-level hierarchy: 8 clusters, each containing 9 regions, each containing 10 routers.

1. Entries for routers in the same region: Each region has 10 routers.

A router needs entries for 10 routers in its region.

2. Entries for other regions in the same cluster: Each cluster has 9 regions.

A router needs 1 entry for each of the other 8 regions in its cluster.

3. Entries for other clusters: There are 8 clusters. A router needs 1 entry for each of the 7 other clusters.



Total Entries:

Total entries = 10 (same region)
 + 8 (other regions) + 7 (other clusters)
 = 25

Q55 Text Solution:

Option- **B(one, seven)**

Explanation: In Bluetooth technology, a piconet is a tiny network composed of one master device and up to 7 slave devices.

- **Master Node:** The central device that manages communication inside the piconet.

- **Slave Nodes:** Devices linked to the master for communication.

Q56 Text Solution:

Option- **B(2,000,000bps)**

Explanation: Given Data:

Bandwidth: 10 Mbps

Frames per minute: 12,000 frames/minute.

Bits per frame: 10,000 bits/frame.

Step 1: Convert frames per minute to frames per second:

$$\begin{aligned}\text{Frames per second} &= \frac{\text{Frames per minute}}{60} \\ &= \frac{12,000}{60} = 200 \text{ frames/second}\end{aligned}$$

Step 2: Calculate throughput:

$$\text{Throughput} = 200 \times 10,000 = 2,000,000 \text{ bps (bits per second).}$$

Q57 Text Solution:

Option- **A(Internet Corporation for Assigned Names and Numbers)**

Explanation- ICANN stands for the Internet Corporation for Assigned Names and Numbers. It is a non-profit organization that coordinates the maintenance and processes of numerous databases connected to the Internet's namespaces and number spaces.

Q58 Text Solution:

Option- **C(Bel(A) + Bel($\neg$ A) = 1)**

Explanation:

The Dempster-Shafer Theory: In the Dempster-Shafer theory for uncertainty management:

- Belief (Bel): The level of belief in a statement (A) based on available evidence.

- Plausibility (Pls): Maximum belief that might support (A), including all uncommitted evidence. The key relationship is: $\text{Bel}(A) + \text{Pls}(\neg A) = 1$

Belief and Complementary Relationship:

The conviction in a proposition (A) and its counterpart ($\neg A$) must satisfy: $\text{Bel}(A) + \text{Bel}(\neg A) = 1$

Evidence supporting (A) and ($\neg A$) may not fully account for all alternatives.

Q59 Text Solution:

Option- **C(Both S_1 and S_2)**

Explanation:

S_1 : These are not equivalent because

$\forall xP(x)\forall xQ(x)$ implies either all ($P(x)$) is true or all ($Q(x)$) is true, whereas $\forall x(P(x)\vee Q(x))$ means for each (x), either ($P(x)$) or ($Q(x)$) is true.

S_2 : $\exists xP(x)\wedge\exists xQ(x)$ implies that ($P(x)$) is true for some (x) and ($Q(x)$) is true for (potentially another) (x), whereas $\exists x(P(x)\wedge Q(x))$ necessitates that both ($P(x)$) and ($Q(x)$) are true for the same (x).

Q60 Text Solution:

Option- **B(17)**

Explanation: To solve the min-max tree question always start from the bottom.

At **level 1** we have to find the minimum of given elements.

From left sub-tree:

$$1. \min(17, 19) = 17$$

$$2. \min(16) = 16$$

$$3. \min(111, 112) = 111$$

Similarly from the right sub-tree:

$$1. \min(15) = 15$$

$$2. \min(14) = 14$$

At **level, 2** we have to find the maximum of the given element present 1.



From left sub-tree:

$$1. \max(17, 16) = 17$$

$$2. \max(111) = 111$$

Similarly from the right sub-tree:

$$1. \max(15, 14) = 15$$

At **level, 3** we have to find the minimum element present at level 2.

From left sub-tree:

$$1. \min(17, 111) = 17$$

From the right sub-tree:

$$1. \min(15) = 15$$

Now at **level, 4** we have to find the maximum of the elements present at level 2.

$$\max(17, 15) = 17$$

Q61 Text Solution:

Correct option is B.

Mamdani model : In a Mamdani system, output of each rule is a fuzzy set. These are well suited to expert system applications where the rules are created from human expert knowledge. It is a method to create a control system by synthesizing a set of linguistic control rules obtained from experienced human operators.

Takagi - Sugeno-Kang model: This model allows to construct the system of analysis from a set of example data. These are used for the approximation of a large class of non - linear systems. The degree of relevance of a linear model is determined by the degree the input data belongs to fuzzy subspace associated with the linear model. This is an example of additive rule model.

Kosko's additive model : It assumes the inputs are crisp and uses the scaling method. It uses addition to combine the conclusion of fuzzy rules and includes the centroid defuzzication technique.

Q62 Text Solution:

The correct option is B.

$$t(x, y) = 1 - s(1 - x, 1 - y)$$

$$t(x, y) = \frac{xy}{(x + y - xy)}$$

$$s(x, y) = 1 - \frac{1 - y - x + xy}{1 - x + 1 - y - (1 - x)(1 - y)}$$

$$s(x, y) = 1 - \frac{1 - y - x + xy}{1 - xy}$$

$$s(x, y) = \frac{1 - xy - 1 + y + x - xy}{1 - xy}$$

$$s(x, y) = \frac{x + y - 2xy}{1 - xy}$$

Q63 Text Solution:

The correct option is B.

The weight updation formula for a multilayer perceptron using the gradient descent method is given by:

$$W_{ij}(t+1) = W_{ij}(t) - \alpha \frac{\partial E}{\partial W_{ij}}$$

Where:

- $W_{ij}(t+1)$ is the updated weight.
- $W_{ij}(t)$ is the current weight.
- α is the learning rate ($0 \leq \alpha \leq 1$).
- $\frac{\partial E}{\partial W_{ij}}$ is the gradient of the error with respect to the weight.
- So option B will be correct.

Q64 Text Solution:

The correct option is B.

The order of schema is the number of fixed positions in a given template.
in the given template ?10?101, the symbol ? is a wildcard symbol which means these places have a value of either 0,1.

- 1) Order schema of ?10?101 = 5 as places have fixed values. (position 2,3,5,6,7)
- 2) Order schema of ???0??1 = 2 as 2 places have fixed values. (position 4,7)

Q65 Text Solution:

The Correct option is D.

$$P = \{(01101), (11000), (01000), (10011)\}$$

Also, the strength of fitness of a chromosome in decimal form, x is given by:

$$Sf(x) = \frac{f(x)}{\sum f(x)} \text{ Where } f(x) = x^2$$

So, first convert the population in decimal form, then find f(x).



| P | Value in decimal | $f(x) = x^2$ |
|-------|------------------|--------------|
| 01101 | 13 | 169 |
| 11000 | 24 | 576 |
| 01000 | 8 | 64 |
| 10011 | 19 | 361 |

We have to find the strength of fitness of chromosome 11000.

11000 in decimal = 24, $f(24) = 576$

$Sf(x) = 576/169+576+64+361$

$Sf(x) = 576/1170$

$Sf(x) = 0.492$

Q66 Text Solution:

The correct option is C.

The first statement is true: As $a \in G$ is such that $a^m = e$ for some integer $m \leq n$, then it means it is a subgroup of G and m is the order of a . where the order of the given group is m . By the property of a subgroup or Lagrange's theorem, an order of subgroup divides the order of a group. So, here m must divide n is true.

The second statement is also true: Consider an example for this: Consider G is of order $2n$. There exists $a \in G$ such that $a^p = e$ and p divides $2n$. Let $n = pq$. So, $(a^n)^2 = (a^{pq})^2 = ((a^p)^q)^2 = (e^q)^2 = e$. It means a^n is an element which satisfies the condition and $a \neq e$. Here, a is a non-trivial subgroup of G .

Q67 Text Solution:

The correct option is D.

Duality in Linear programming problem has another LPP which is derived from it. Original LPP is known as primal and derived LPP is known as Dual.

Some points about dual LPP:

- 1) The final simplex table giving the optimal solution of the primal also contains the optimal solution of its dual in itself.
- 2) If either the primal or the dual problem has a finite optimal solution, then the other problem also has a finite optimal solution.
- 3) If either problem has an unbounded optimum solution, then the other problem has no feasible

solution at all.

Q68 Text Solution:

The correct option is C.

Statement A is correct: RISC has pipelined implementations with the goal of executing one instruction per machine cycle.

Statement B is incorrect: RISC has fixed-length instruction formats.

Statement C is incorrect: RISC can perform only Register to Register Arithmetic operations.

Statement D is correct: RISC has efficient instruction pipeline.

So, only statements a and d are correct.

Q69 Text Solution:

The correct option is B.

Self-complementing codes are: excess-3 code, and 2421 code.

1) 2421 code: It is positive weighted code Values in binary terms for each digit from 0 to 9 are given as:

| Decimal digit | 2421 code |
|---------------|-----------|
| 0 | 0000 |
| 1 | 0001 |
| 2 | 0010 |
| 3 | 0011 |
| 4 | 0100 |
| 5 | 1011 |
| 6 | 1100 |
| 7 | 1101 |
| 8 | 1110 |
| 9 | 1111 |

2) Excess-3 code: It is a non weighted code and self-complementary code. It is derived from 8421 code by adding 3 to each digit.

| Decimal digit | Excess-3 code |
|---------------|---------------|
|---------------|---------------|



| | |
|---|----------|
| 0 | 0011 |
| 1 | 010
0 |
| 2 | 0101 |
| 3 | 0110 |
| 4 | 0111 |
| 5 | 100
0 |
| 6 | 1001 |
| 7 | 1010 |
| 8 | 1011 |
| 9 | 1100 |

Q70 Text Solution:

The correct option is C.

There are two approaches used in the area filling on raster systems: one is the scan line algorithm and another is the seed fill algorithm.

1) Scan line algorithm: It determines the overlap intervals for scan lines that cross the area or scan line. A scan line is a line of pixels that are plotted on a display device when an electron beam scans the display area horizontally.

2) Seed fill algorithm: It starts from the interior position and points outward from this point until the boundary condition is reached. In this, a point is selected and then four or eight connected approaches is used to fill it.

Q71 Text Solution:

The correct option is C.

Overloading gives the capacity to an existing operator to operate on other data types. True
Inheritance in object-oriented programming provides support to reusability. True

When an object of a derived class is defined, first the constructor of the derived class is executed then the constructor of a base class is executed.

True

Overloading is a type of polymorphism. True

2 important points Order of Constructor Call with Inheritance in C++

Whether the derived class's default constructor is called or parameterized is called, the base class's

default constructor is always called inside them. To call the base class's parameterized constructor inside the derived class's parameterized constructor, we must mention it explicitly while declaring the derived class's parameterized constructor.

Q72 Text Solution:

The correct option is C.

`int *P=&44;` → assigns the address of constant to a variable.

`int P=&a;` → assigns the address of a variable to a variable of type `int`.

Both the statements are not valid.

Q73 Text Solution:

The correct option is D.

Problems that can occur during the concurrent execution of two transactions in an uncontrolled manner.

a) Lost update problem: A lost update problem occurs when two transactions that access the same database items have their operations interleaved in a way that makes the value of some database item incorrect.

b) Dirty read problem: It occurs when a transaction is allowed to retrieve or update a record that has been updated by another transaction. Because it has not yet been committed, there is always a possibility that it will never be committed but rather roll back.

d) Inconsistent database state: It occurs when a transaction reads several values but a second transaction updates some of these values during the execution of the first. This causes an inconsistency.

Q74 Text Solution:

The correct option is A.

CPU scheduling decisions take place under one of four conditions:

1. When a process switches from the running state to the waiting state, such as for an I/O request or invocation of the `wait()` system call.
2. When a process switches from the running state to the ready state, for example in response



to an interrupt.

3. When a process switches from the waiting state to the ready state, say at completion of I/O or a return from wait().

4. When a process terminates.

For conditions 1 and 4 there is no choice - A new process must be selected.

For conditions 2 and 3 there is a choice - To either continue running the current process, or select a different one.

If scheduling takes place only under conditions 1 and 4, the system is said to be non-preemptive, or cooperative. Under these conditions, once a process starts running it keeps running, until it either voluntarily blocks or until it finishes.

Otherwise, the system is said to be preemptive. In above discussion, it's given that If scheduling takes place only under conditions 1 and 4, the system is said to be non-preemptive. Otherwise, the system is said to be preemptive.

Therefore point 2 & 3 are condition for preemptive system. Here point 2 and 3 are as listed below When a process switches from the running state to the ready state. When a process switches from the waiting state to the ready state.

These above point matches with circumstances (a) & (b) given in question.

Q75 Text Solution:

The correct option is A.

During the transpose of the matrix, it is partitioned into 4 submatrices of size $n/2$.

If $n = 1$, then $T(1) = 1$

Else $T_1(n) = 4T_1(N/2) + O(1) = O(N^2)$ {Using Master's theorem}

$T_\infty(N) = T_\infty(N/2) + O(1) = O(\log N)$ {Using Master's theorem}

The asymptotic parallelism of the algorithm = $\frac{T_1}{T_\infty} = \left(\frac{N^2}{\log N}\right)$

Q76 Text Solution:

The correct option is B.

Statement (i) is false: The running time depends on the number of subproblems multiplied by the

time taken by each subproblem.

It will be true only in the case when the time taken by each subproblem = $O(1)$.

Statement (ii) is true: Cyclic dependency has mutually recursive modules. So, we cannot find the correct solution in DP.

Statement (iii) is false: The reverse of the given statement is true, i.e. using Recursion and memoization it is faster.

Statement (iv) X must be NP-complete and not NP-hard.

Q77 Text Solution:

The correct option is A.

(a) Fiber optic cable is much lighter than copper cable Given statement is correct. Fibre optic cable transmit signal in the form of light pulses. They can transmit data at higher speed than copper cables and these are lighter than copper cables. Also, fibre optic cable have higher bandwidth.

(b) Fiber optic cable is not affected by power surges or electromagnetic interference This statement is correct. Fibre optic cables are non-metallic and they transmit signals in the form of light pulses. That's why they are immune to electromagnetic interference. The integrity of the signal is not affected by electrical noise.

(c) Optical transmission is inherently bidirectional. This statement is incorrect. Two optical modules are connected by a single fibre cable which supports unidirectional communication. For bi-directional communication, two cables will be required.

Q78 Text Solution:

The correct option is B.

(a) Windows Azure is a cloud-based operating system. This statement is correct. Windows Azure is a cloud-based operating system which enables running of business applications, services and workload in the cloud itself. It mainly deals with the foundation of running applications and keeping the data on the cloud.

(b) Google App Engine is an integrated set of online services for consumers to communicate



and share with others. Google app engine is PaaS (Platform as a service) product. It provides web developing services to developers. Apps must be written in java or python. So, this statement is incorrect.

(c) Amazon Cloud Front is a web service content delivery. Amazon cloud front is a fast content delivery network service that securely delivers data, video, applications and APIs to customers. Cloud front is integrated with AWS global infrastructure and services. So, statement is correct.

Q79 Text Solution:

The correct option is D.

- Message Confidentiality- It means a set of rules and promises that are executed through confidentiality agreements.
- Message Integrity - It means that the data must not be modified in transit.
- Message Authentication - It is the property that the message is not changed and also the source can be verified by the receiver.

Q80 Text Solution:

The correct option is D.

Problems in hill climbing are:

- 1) Local maximum: It is a state that is better than all of its neighbors but is not better than some other states that are farther away. The solution of this problem is backtracking.
- 2) Plateau: It is a flat area of search space in which a whole set of neighbouring states have the same value. On a plateau, it is not possible to determine the best direction in which to move.
- 3) Ridge: It is an area of search space which is higher than the surrounding areas and that itself has a slope.

Q81 Text Solution:

Option A is correct

Logistic Regression:

Logistic regression is a classification algorithm used to predict discrete outcomes (e.g., binary classes such as 0/1, spam/not spam). It uses a logistic function (sigmoid) to map predictions to probabilities. Hence, it can be categorised as Classification algorithm.

Backpropagation:

Backpropagation is a learning algorithm used in training neural networks. Neural networks are versatile and can be used for both classification and regression tasks. It is a Classification algorithm when applied to classification problems (e.g., image recognition).

Linear Regression:

Linear regression is a regression algorithm used to predict continuous outcomes (e.g., house prices, temperature). It fits a linear relationship between input variables and the output. Hence, It is Not a classification algorithm.

Q82 Text Solution:

Option C is Correct

| List I | List II | Explanation |
|------------------------|---|--|
| (a) Physical Layer | (ii) Concerned with transmitting raw bits | Responsible for transmitting raw bits over a communication channel. |
| (b) Transport Layer | (iv) True end-to-end layer | Provides reliable, end-to-end communication between source and destination. |
| (c) Session Layer | (i) Provide token management service | Manages sessions and provides token management services for communication. |
| (d) Presentation Layer | (iii) Concerned with syntax and semantics | Concerned with the syntax and semantics of the information being transmitted (e.g., data encoding, compression). |

Q83 Text Solution:

Option C is Correct



| List I | Matched with List II | Explanation |
|---------------------|--|---|
| (a) Functionality | (ii) Characteristics related with achievement of Purpose | Functionality is the software's ability to achieve its intended purpose. It focuses on essential features provided. |
| (b) Reliability | (iv) Capability of software to maintain performance of software | Reliability ensures the software maintains its performance, even under failure conditions, such as fault tolerance. |
| (c) Efficiency | (i) Relationship between level of performance and amount of resource | Efficiency concerns the relationship between the software's performance and the resources (disk, memory, network) used. |
| (d) Maintainability | (iii) Effort needed to make for improvement | Maintainability relates to the ease of identifying and fixing faults or making improvements in the software. |

Q84 Text Solution:

Option A is Correct

| List I | Matched with List II | Explanation |
|-----------------------------------|----------------------------|--|
| (a) Scrum | (ii) Sprint backlog | Scrum uses a sprint backlog to manage tasks that need to be completed within a time-boxed sprint. |
| (b) Adaptive software development | (iv) Time box release plan | Adaptive software development focuses on iterative cycles, with tasks planned using a time-box release plan. |
| (c) Extreme programming (XP) | (i) CRC cards | XP uses CRC (Class Responsibility Collaborator) cards for designing and identifying classes. |
| (d) Feature- | (iii) the <by/for/ | Feature-driven development emphasizes tasks structured |

| List I | Matched with List II | Explanation |
|--------------------|----------------------|---|
| driven development | of/to> a(n) | as <by/for/of/to> statements to deliver features. |

Q85 Text Solution:

Option C is Correct

| List I | Matched with List II | Explanation |
|----------------------|---|---|
| (a) Frame attribute | (iv) to specify the side of the table frame that display border | The frame attribute specifies the side(s) of the table frame where borders should be displayed. |
| (b) <tr> tag | (iii) to enclose each row in table | The <tr> tag is used to define a table row, enclosing all the cells in that row. |
| (c) Valign attribute | (ii) for vertical alignment of content in cell | The valign attribute controls the vertical alignment of content within a table cell. |
| (d) <a> tag | (i) to create link in html | The <a> tag is used to create a hyperlink in HTML, linking to other resources or pages. |

Q86 Text Solution:

Option A is Correct

| List I (Addressing Mode) | Matched with List II (Effective Address) | Explanation |
|--------------------------|--|--|
| (a) Direct | (iii) 400 | The address field directly contains the effective address, which is 400. |
| (b) Register Indirect | (i) 200 | The register R1 contains the effective address directly, which is 200. |
| (c) Index with R1 | (iv) 600 | Effective address is calculated as |



| List I
(Addressing Mode) | Matched with List II
(Effective Address) | Explanation |
|-----------------------------|---|--|
| | | address field+R1=400+200=600. |
| (d) Relative | (ii) 902 | Effective address is calculated as next instruction address+address field=502+400=902. |

Q87 Text Solution:

Option C is Correct

| List I | List II | Explanation |
|--------------------------------|--|---|
| (a) Isolated I/O | (ii) Separate instructions for I/O and memory communication | Isolated I/O uses specific instructions (e.g., IN, OUT) for I/O operations distinct from memory access. |
| (b) Memory-mapped I/O | (i) Same set of control signals for I/O and memory communication | Memory-mapped I/O treats I/O devices as memory locations, using the same control signals for both. |
| (c) I/O interface | (iv) Resolve the differences in central computer and peripherals | The I/O interface handles differences in speed, data format, and signaling between the CPU and peripherals. |
| (d) Asynchronous data transfer | (iii) Requires control signals to be transmitted between the communicating units | Asynchronous data transfer depends on control signals to synchronize communication between devices. |

Q88 Text Solution:

Option A is Correct

| List I | Matched with List II | Explanation |
|---------------------|---------------------------|---|
| (a) Micro operation | (iv) Elementary operation | Micro operations are low-level operations like data transfer, arithmetic, |

| List I | Matched with List II | Explanation |
|----------------------------------|---------------------------------------|--|
| | performed on data stored in registers | or logic performed on registers. |
| (b) Microprogrammed control unit | (iii) Control Memory | Microprogrammed control units use control memory to store microinstructions for generating control signals. |
| (c) Interrupts | (ii) Improve CPU utilization | Interrupts allow the CPU to handle critical tasks, improving its overall utilization by avoiding idle time. |
| (d) Micro instruction | (i) Specify micro operations | Micro instructions are small instructions within the microprogram that specify the sequence of micro operations. |

Q89 Text Solution:

Option C is Correct

The order reflects the objects involved in the reply chain, beginning with the source of information (Weather Data) and ending with the final recipient (Sat Comms). This matches the sequence given in the detailed explanation.

Q90 Text Solution:

Option B is Correct

The Data Encryption Standard (DES) has a specific sequence of according to which The correct order of steps in the DES function is:

1. Expansion Permutation:

Expands the 32-bit input into 48 bits by duplicating certain bits.

2. XOR Operation:

Performs an XOR between the expanded 48-bit block and the 48-bit subkey.

3. S-Boxes (Substitution Boxes):

Substitutes the XOR output (48 bits) using eight S-Boxes to reduce it back to 32 bits.

4. Straight Permutation:



Applies a fixed permutation on the 32-bit output from the S-Boxes to provide diffusion.

Q91. Text Solution:

Option A is Correct
Nodes in the Graph

1. 1
2. (2,3) (treated as one combined node)
3. (4,5) (treated as one combined node)
4. 6
5. 7
6. 8
7. 9
8. 10
9. 11

Q92. Text Solution:

Option C is Correct
According to given Graph
Total Number of Vertex (V) = 9
Total Number of Edges (E) = 11
Number of Predicate Node = 3 $\rightarrow$ (1,(2,3),6)
So, The Cyclomatic Complexity of flow graph =
number of predicate node + 1 = 3+1 = 4

Q93. Text Solution:

Option C is Correct
A region is an area that is bounded by edges and cannot be further subdivided. Region can be finite nature and can also be infinite nature. A planar graph can only have one infinite region. There are 4 Regions

- (1, (2,3), (4,5), 10)
- 6,7,8,9
- 1, (2,3), 6,8,9,10, (4,5)
- Outer Side Region

Q94. Text Solution:

Option C is Correct
A linearly independent path introduces at least one new edge that is not included in any previously defined independent path. It does not repeat edges unnecessarily.
The identified independent paths are:

1. Path 1: 1 $\rightarrow$ 11

2. Path 2: 1 $\rightarrow$ 2 $\rightarrow$ 3 $\rightarrow$ 4 $\rightarrow$ 5 $\rightarrow$ 10 $\rightarrow$ 1 $\rightarrow$ 11
3. Path 3: 1 $\rightarrow$ (2,3) $\rightarrow$ 6 $\rightarrow$ 8 $\rightarrow$ 9 $\rightarrow$ 10 $\rightarrow$ 1 $\rightarrow$ 11
4. Path 4: 1 $\rightarrow$ (2,3) $\rightarrow$ 6 $\rightarrow$ 7 $\rightarrow$ 9 $\rightarrow$ 10 $\rightarrow$ 1 $\rightarrow$ 11

Among the paths, Path 3 and Path 4 include 8 nodes, which are the longest independent paths.

Q95. Text Solution:

Option A is Correct
From the graph, the nodes with **two or more outgoing edges** are:

1. **Node 1:** Splits into multiple paths.
2. **Node (2,3):** Acts as a single decision point with multiple outgoing edges.
3. **Node 6:** Splits into multiple paths.

Thus, there are **3 predicate nodes**: 1, (2,3), and 6.

Q96. Text Solution:

Option A is Correct
Given Functional Dependency Set F:
 $F = \{A \rightarrow BC, D \rightarrow E, BC \rightarrow D, A \rightarrow D\}$
Steps to Find Minimal Cover:

1. Break Down Multi-Valued Dependencies:
 - $A \rightarrow BC$ can be split into:
 - $A \rightarrow B$
 - $A \rightarrow C$
2. Remove Redundant Dependencies:
 - $A \rightarrow D$ is redundant because:
 - From $A \rightarrow BC$ ($A \rightarrow B$ and $A \rightarrow C$) and $BC \rightarrow D$, we can infer $A \rightarrow D$.
3. Minimal Cover (F'):
 - $F' = \{A \rightarrow B, A \rightarrow C, BC \rightarrow D, D \rightarrow E\}$

Q97. Text Solution:

Option C is Correct

1. Closure of A:

- $A^+ = A \rightarrow BC$
- $A^+ = A \rightarrow B \rightarrow C \rightarrow D$
- $A^+ = A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$
- A^+ includes all attributes (A,B,C,D,E).

2. Essential Attribute:



Since A can determine all attributes, it is the only candidate key. Hence, It will become Primary Key

Q98. Text Solution:

Option B is Correct

Given Relation

schema $R(ABCDE)$ and FD set $\{A \rightarrow BC, D \rightarrow E, BC \rightarrow D, A \rightarrow D\}$

Now, we can find the Candidate Key(s) of the relation R using the FD set and the Candidate key of the relation R will be A.

If some attribute(s) are not on the right side of any non-trivial FD, then those attribute(s) must be the part of every Candidate Key.

These attribute(s) are called essential attribute(s).

In the given problem, A alone is essential attribute and it can find all other attributes of the relation R. If essential attribute(s) themselves can find all other attributes, then only single Candidate Key exists for a relation. So, A is the only Candidate Key of the given relation R.

To prove this, we will find closure of A –

$A^+ = ABCD$ (as $A \rightarrow BC$ and $A \rightarrow D$)

$A^+ = ABCDE$ (as $D \rightarrow E$)

Checking for 2NF:

Since, the Candidate key is having single attribute, so it cannot derive any Non-prime attribute. Hence, No Partial Dependency exists.

Hence, the Relation R is in 2NF.

Checking for 3NF:

Look at the Non-trivial FD $BC \rightarrow D$. It is a Transitive dependency as the non-prime attributes(B, C) is deriving other Non-key attribute D. Hence, We can say that Non-prime attribute D is Transitively

dependent on candidate key A. So, there is Transitive Dependency. Hence, the Relation R is not in 3NF.

Q99. Text Solution:

Option C is Correct

Redundancy Check:

$A \rightarrow D$:

- From $A \rightarrow BC$ (which can be split into $A \rightarrow B$ and $A \rightarrow C$) and $BC \rightarrow D$, we can infer $A \rightarrow D$.
- Thus, $A \rightarrow D$ is redundant.

Other Dependencies:

$BC \rightarrow D$, $D \rightarrow E$, and $A \rightarrow BC$ are essential for the minimal cover.

Q100. Text Solution:

Option 4 is Correct

Relation $R(A, B, C, D, E)$ with its functional dependencies $F : (A \rightarrow BC, D \rightarrow E, BC \rightarrow D, A \rightarrow D)$ has been decomposed in two relations.

1. $R_1(A, B, C)$ with functional dependency set $f_1 = \{A \rightarrow B, A \rightarrow C\}$

2. $R_2(A, D, E)$ with FD set $F_2 = \{A \rightarrow D, D \rightarrow E\}$

In both the relations, the dependency $BC \rightarrow D$ cannot be determined. Hence, both R_1 and R_2 are not following Dependency Preservation property.

Normal Forms In R_1 , A is the candidate key. And the functional dependency set does not contain any transitive partial dependency.

Hence, the set is in 3NF.

In R_2 , A is the candidate key. But the FD $D \rightarrow E$ breaks the rule of 3NF. Hence, it is in 2NF.

